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The Effect of Innovation Similarity on Asset Prices: Evidence from Patents' Big Data

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Through textual analyses of 7.7 million patents, we develop a novel intercompany innovation similarity measure which enables us to find that technologically connected firms cross-predict one another's returns. Investors impound information about firms' technological connectedness, although not immediately and fully. Buying (shorting) shares of technological peers earning high (low) returns during the previous month yields a 1.29% monthly return. Firms' return predictability increases with patent complexity or limited technological disclosures but decreases with better information transparency. Results suggest that investor inattention explains technology momentum. Unlike momentum stemming from simpler, class-based technological links, our Big Data text-based return predictability remains active. (*JEL* G11, G12, G14, O31, C55)

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Economists have long agreed that technological innovation activity is the main engine for sustained economic growth (Solow 1957; Romer 1990; Aghion and Howitt 1990; Klette and Kortum 2004; Grossman and Helpman 1991). Indeed, effective innovations continue to improve firm performance for several years after they are generated (Hall, Jaffe, and

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Trajtenberg 2005). As a result, spillovers as well as competition in the technological space among innovative firms can determine their success. In this paper, we develop a novel measure of innovation similarity among firms and use it to investigate how investors impound information about cross-firm technological links in stock prices.

To identify technologically similar firms, we use the publicly available and extremely rich information contained in patent descriptions. We leverage the latest advancements in Big Data science to automate the analysis of approximately 400 gigabytes of textual data from 7.7 million patent descriptions, creating a unique continuous patent-to-patent similarity space in which topically similar patents are connected. Then we aggregate individual patents at the firm level to describe a firm's innovation space. This process allows us to construct a novel patent-to-patent measure of technological distance among firms.

In theory, a firm's stock price will quickly adjust to reflect material information about a close technological peer when such information becomes public. However, if, due to limited processing capacity or inattention, investors do not immediately and fully incorporate the arrival of relevant public information, the stock prices of related firms will have a predictable lag in reacting to new information about technologically linked firms. We hypothesize that investors will have difficulty processing information about innovation similarity because of its technicality, uncertainty, and content specificity (Lev and Sougiannis 1996; Chan, Lakonishok, and Sougiannis 2001; Farboodi and Veldkamp 2020). Therefore, we test the asset-pricing implications of technological links. To do so, for each focal firm we estimate the portfolio return for its text-based technologically linked firms, weighted by the strength of their technological similarity to the focal firm.

To better understand our approach, consider the events of October 2015 and their impact on the stock prices of Amazon.com and its technological peers. October 6–9, 2015, Amazon held the largest annual cloud computing conference (re:Invent). During the 5-day window following the conference, Amazon's cumulative abnormal return (CAR) reached 5.08%.[1] Additionally, in their October 22nd earnings announcement, Amazon provided strong evidence of astonishing growth in the Amazon Web Services (AWS) segment: sales reached $2.1 billion, which implied a 78% growth over the same quarter of the previous year. Amazon's 5-day CAR after the AWS announcement was 9.3%. These events account for Amazon's abnormal return of 14.87% for the entire month of October.

The good news disclosed by Amazon also increased the value of its technological neighbors' stocks, as the rapid expansion of cloud computing is beneficial to the entire cloud ecosystem. Figure 1 shows the CARs for

[1] The 5-day CAR is calculated as the 5-day cumulative return adjusted by the Center for Research in Security Prices' (CRSP's) value-weighted average return.

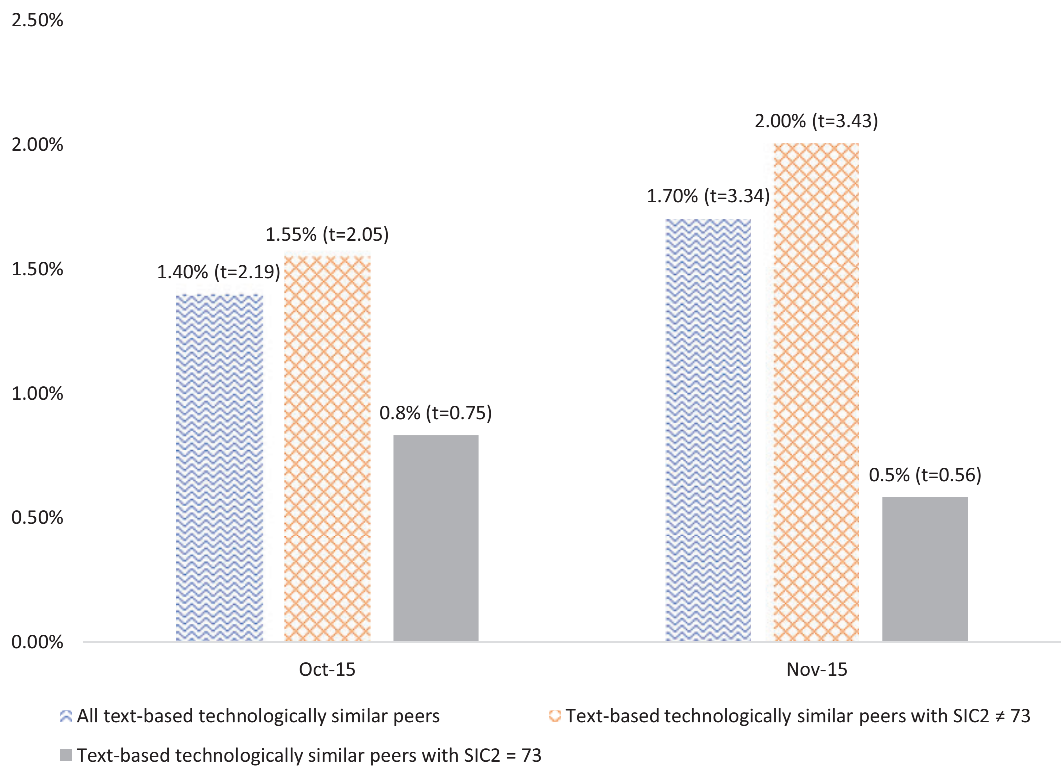

Figure 1
Cumulative abnormal return of Amazon.com's technological peers
This figure shows the cumulative abnormal returns of Amazon.com's technological neighbors in October 2015, when a few events that significantly impacted stock prices of Amazon and its technological peers occurred, and November 2015. October 6–9, 2015, Amazon held the largest annual cloud computing conference, re:Invent. On October 22, 2015, at their earnings announcement, Amazon provided strong evidence of astonishing growth in the Amazon Web Services segment.

Amazon's technological neighbors in October and November of 2015. Indicating a significant return comovement, the CAR for all of Amazon's technological peers was 1.39% (t-statistic = 2.19) for the month of the announcements (October 2015). Moreover, showing a significant lead–lag effect, the CAR for all technological peers was 1.70% (t-statistic = 3.33) in the following month (November 2015). Importantly, industry links do not affect the contemporaneous comovement and lead–lag effect for Amazon's technological peers. The CAR for technological peer firms that do not belong to the same two-digit Standard Industrial Classification (SIC) industry as Amazon was even higher: 1.55% (t-statistic = 2.05) in October, and 2% (t-statistic = 3.43) in November.

Consistent with the Amazon case study, our empirical analyses document a similar pattern that is systematic for the cross section of U.S. common stocks. Specifically, we find that investors do incorporate ex ante publicly available information about technological connections in stock prices and confirm that the stock returns of text-based technologically similar firms comove contemporaneously. However, investors do not incorporate this information immediately and fully: the monthly stock return of a focal firm is predicted by the lagged portfolio of text-based technologically similar firms.

We show that self-financing trading strategies based on the cross-predictability of technologically similar firms yield economically and statistically significant returns. A trading strategy consisting of buying stocks of firms with technologically similar peers with high returns during the previous month and shorting stocks with low peer returns during the same period yields a 1.29% monthly return (17% annualized). The returns on these strategies are not explained by the widely used risk factors (Fama and French 1993, 2015) and remain profitable over the sample period.

To rule out the possibility that our results are driven by the previously documented industry momentum (Hou and Moskowitz 2005; Hou 2007; Moskowitz and Grinblatt 1999), we rigorously control for product market similarity by including two industry portfolio returns: (i) the value-weighted Fama and French (1997) 49 industry portfolios and (ii) Hoberg and Phillips's (2010, 2016) dynamic text-based network industry classification (TNIC) portfolio, weighted by the strength of the product market similarity. We also control for short-term reversals and midterm price momentum, various cross-sectional characteristics, and the stock returns of firms linked by the traditional measures of technological similarity (e.g., citation overlap and the U.S. Patent and Trademark Office [USPTO] classification). None of these controls alters our results.

Since we develop a novel text-based measure of innovation similarity or proximity, we perform several tests aimed at validating it. The results of these analyses show that the new measure can predict future patent citations by technologically linked firms and serves as a forward-looking indicator, predicting product market similarity in the following years. We also find that a firm's profitability, measured by return on assets (ROA), is correlated with the contemporaneous and lagged weighted average ROA of technologically connected peers.[2] Additionally, a firm's innovation activity, measured by R&D intensity, is correlated and has a lead–lag relation with the weighted average R&D intensity of linked firms.[3] Moreover, we show that our text-based measure of innovation proximity predicts the likelihood of forming a mergers and acquisitions (M&A) pair. This result extends Bena and Li's (2014) findings that firms are more likely to merge if they exhibit technological overlap derived from either patent classification or reciprocal patent citations. Collectively, our validation results indicate that the technological peers identified by our Big Data text-based measure are indeed peers. Importantly, the ROA and R&D results also cast doubt on the alternative that some unobservable risk factor might drive our results.

[2] To ensure that our results are not driven by characteristics of the economic cycle, industry, or firm, we control for a firm's own lagged profitability, R&D intensity, size, and book-to-market ratio as well as year and industry fixed effects.

[3] The weighted average ROA (R&D intensity) is the average ROA (R&D intensity) of the portfolios of technologically similar firms, where each firm is weighted by the strength of its technological similarity to the focal firm. Our results remain qualitatively similar if we use equal-weighted measures.

We perform a battery of robustness checks to ensure that our asset pricing results are not driven by the limited liquidity of the small stocks or by the previously documented lead–lag relations between large and small firms (Hou 2007). To mitigate concerns about limited liquidity, we obtain a baseline result for the sample that excludes stocks with a price below \$5. We then repeat the analysis imposing additional restrictions on our sample. First, we exclude firms with a market capitalization below the 20th percentile of the New York Stock Exchange (NYSE). Second, we drop all links when a peer firm is larger than the predicted firm. To rule out the possibility that customer–supplier links drive our results (Menzly and Ozbas 2010; Cohen and Frazzini 2008), we control for the portfolio return of customers' industries. After we perform all of these robustness tests, the tenor of our findings remains unchanged.

The closest study to ours is contemporaneous work by Lee, Sun, Wang, and Zhang (2019), hereafter, LSWZ. They identify peer firms by checking whether their innovations fall under the same patent class. Using this taxonomy, LSWZ find that a firm's stock return is predicted by the returns of its technologically linked peers. They rule out industry momentum and risk-based explanations for their results. Consequently, LSWZ also interpret this evidence as consistent with the hypothesis that limited investor attention explains the predictability of stock returns. Given that some of our results overlap with those of LSWZ, we perform additional analyses that differentiate this study from theirs and expand the contributions of our work.

Using a sample of firms with stock prices above \$1, LSWZ find return predictability during their whole sample period, with markedly smaller alphas during 2000–2012. This annual secular decrease in their reported alpha could be due to a phenomenon analogous to the one studied by McLean and Pontiff (2016) whereby the return to anomalies fades when they become public. Consistent with this conjecture, we show that for a sample of firms with stock prices above \$5, the patent class-based return predictability of LSWZ vanishes after 2000. Conversely, our text-based measure exhibits return predictability throughout the entire sample period. Moreover, the predictability pattern of the "pure" text-based technological peers (those with no overlap with patent class-based peers) highlights how investors' practice of incorporating patent information has evolved. "Pure" text-based peers do not show return predictability before 1999 when patents' text was not available through online access. However, with the advent of online access, investors could examine patent text and draw conclusions about firms' technological positions. In line with this conjecture, we find that "pure" text-based peers exhibit strong predictability from 2000 onwards.[4] Notably, our text-based approach is

[4] In 1996, Congress amended the Freedom of Information Act (FOIA) statute to require agencies to create electronic reading rooms for records that are likely to become FOIA requests. In 1999, the public gained access to images of all patents issued from 1976 to the present.

immune to the issues associated with the use of patent classes to determine technological similarity. Indeed, as noted by Thompson and Fox-Kean (2005, 453–55) the USPTO selects the primary class as the "broadest claim" in a patent, which might miss the main thrust of the innovation that originated the patent. Thus, in contrast with the class-based approach, our text-based methods better capture the main thrust of the innovation to establish technological connections among firms. These are important differences between our work and LSWZ's. Moreover, future work could build on ours because, unlike LSWZ's, the stock return predictability anomaly related to our text-based measure remains active and strong.

We show that firms with patents that are more complex exhibit stronger predictability. In this test, we determine complexity with the number of keyword phrases in the patents, which we then aggregate at the firm level. Likewise, we find stronger predictability for firms with a higher fraction of "exploitative" patents. In this test, we define patents as exploitative if at least 80% of their technological connections are with their own (previous) patents. The intuition behind the latter finding is as follows. If most of a firm's patent connections are "internal," the information flow related to its innovation activity is limited. Thus, the spillover effect is less obvious for market participants whenever a firm's innovations are built upon (exploit) its previous inventions.

Together, our complexity and exploitative results advance our understanding regarding the features that are difficult for investors to process. In this vein, these findings illuminate the potential channels underlying our baseline results. It is also important to stress that it would be hard (and perhaps impossible) to use the class-based method of LSWZ to classify patents as either complex or exploitative. Therefore, the complexity and exploitative results not only deliver further support for the inattention processing/capacity hypothesis, but also further distinguish our work from LSWZ's by highlighting that our textual analyses methods can identify economically important characteristics that the class-based method cannot.

The contributions of this study are as follows. First, we advance the growing body of research that uses textual analysis to address economic, financial, and product market theories (Hoberg and Phillips 2010; Hanley and Hoberg 2010; Tetlock 2007; Tetlock, Saar-Tsechansky, and Macskassy 2008; Loughran and McDonald 2011, 2016; Currie, Kleven, and Zwiers 2020). We construct a novel patent-to-patent measure of innovation similarity by using the publicly available and extremely rich Big Data of patent text. Traditional measures of a firm's technological position, which are often based on citation overlap or USPTO classification code, have been criticized for their subjectivity, inadequacy of the patent taxonomy, and possible manipulability attributable to patent attorneys and firms' disclosure strategies. We believe that our automated text-based measure of technological proximity is immune to these concerns. While prior research has attempted to improve similarity and distance measures of patent classification (Benner and Waldfogel 2008; Bar and Leiponen 2012; McNamee 2013), their starting point has still been the existing static USPTO patent classification.[5] We built a fully automatic, parameter-free patent mining infrastructure for researchers and practitioners to explore and reexamine numerous important issues in economics, finance, and management such as knowledge and geographical spillovers (Jaffe, Trajtenberg, and Henderson 1993; Thompson and Fox-Kean 2005), inter-organizational collaboration (Gilsing et al. 2008; Bloom, Schankerman, and Van Reenen 2013), corporate diversification (Silverman 1999), and market competition (Teece 1992).

Second, building on the misvaluation of innovation, limited investor attention, and information diffusion literature (Hirshleifer and Teoh 2003; Hirshleifer, Lim, and Teoh 2009; Peng and Xiong 2006; Hong and Stein 1999; Ottaviani and Sørensen 2015; Hou 2007; Aobdia, Caskey, and Ozel 2014; Cao, Chordia, and Lin 2016), we show that not only a firm's own innovations but also the innovations of its technological peers provide important information for its valuation. Using our unique Big Data text-based measure, we show that the market impounds information about innovation similarity into stock prices but fails to do so immediately and completely. We show not only that text-based technologically linked peers predict stock returns beyond industry momentum, but also that smaller stocks can predict larger stocks. Notably, text-based innovation similarity does not seem to decrease over time. Our work adds to the debate about the role of investors' limited cognitive processing capacity, suggesting that investors have greater difficulty processing intangible innovation similarity information than more obvious product similarity information.

1. Background and Literature Review

This section briefly reviews the literature relevant to the research question that we study to both provide a conceptual framework and highlight our contributions.

1.1 Valuation of innovation

The extant literature delivers ample evidence that various indicators of innovation activity, such as R&D intensity and growth (Chan et al. 2001; Penman and Zhang 2002; Daniel and Titman 2006; Garleanu, Panageas, and Yu 2012; Trajtenberg 1990), patent issuance and citations (Gu 2005; Matolcsy

[5] In a concurrent working paper, Younge and Kuhn (2016) also use patent text of the USPTO database to calculate patent-to-patent proximity. They show that a text-based similarity measure improves upon existing patent classification.

and Wyatt 2008, Chari, Golosov, and Tsyvinski 2012), innovation efficiency (Hirshleifer, Hsu, and Li 2013), and originality (Hirshleifer, Hsu, and Li 2017), are associated with strong future operating performance and high stock returns. At the same time, debate continues in the literature regarding whether observed return predictability is attributable to a higher exposure by R&D-intensive firms to systematic risk (Chambers et al. 2002; Li 2011) or to investors' limited ability to process relevant public information. We contribute to this debate by examining the asset pricing implications of innovations undertaken by technologically similar peers.

1.2 Limited attention and information-processing capacity

Over the past 2 decades, a large body of work has gradually identified the role of investor psychology in determining asset prices.[6] Indeed, many financial economists characterize investors as being only boundedly rational, recognizing that investors do not possess unlimited processing capacity (Sims 2006; Barber and Odean 2008; DellaVigna and Pollet 2009; Huberman and Regev 2001). Theoretical work by Merton (1987), later developed by Hong and Stein (1999, 2007), Hirshleifer and Teoh (2003), and Peng and Xiong (2006), outlines how investors' limited attention and processing capacity can lead to return predictability. The rationale underpinning these theories is as follows: if investors consider ex ante publicly available information about economic links, the prices of linked firms will adjust to the information released by similar firms. In contrast, if, due to inattention, investors do not quickly and completely impound the arrival of relevant public information, the stock prices of related firms will have a predictable lag in reacting to new information about connected firms. Motivated by this theoretical literature, recent empirical studies have explored inattention to economic links between firms. In this vein, Hou (2007) and Hoberg and Phillips (2016) find within-industry stock return predictability. Cohen and Frazzini (2008) and Menzly and Ozbas (2010) document the cross-predictability of stock returns along customer–supplier links, and Scherbina and Schlusche (2015) show the return predictability of economic links inferred from news stories. In this paper, we move this literature forward by studying how investors incorporate information on innovation links into stock prices. Drawing from the limited-attention literature, we expect that information about these links is initially only partially incorporated into stock prices and is only fully recognized over time.

1.3 Patent information and a firm's technological position

Patents are often used as a proxy for the output of a firm's innovation efforts (Hall and Ziedonis 2001; Hall, Jaffe, and Trajtenberg 2005; Agrawal and

[6] For a summary of the literature, see Hirshleifer (2001) and Barberis and Thaler (2003).

Henderson 2002; Jaffe, Trajtenberg, and Forgarty 2000; Gu 2005; Matolcsy and Wyatt 2008; Pandit, Wasley, and Zach 2011; Hirshleifer, Hsu, and Li 2013). Researchers have long used patent information to characterize firms' technological positions and estimate their proximity in technological space (Jaffe 1989; Stuart and Podolny 1996; Rosenkopf and Almeida 2003).[7] Technological proximity helps uncover knowledge spillovers (Jaffe, Trajtenberg, and Henderson 1993), knowledge flow between firms (Song, Almeida, and Wu 2003; Mowery, Oxley, and Silverman 1996), and diversification (Silverman 1999; Miller 2004; Argyres 1996; Piscitello 2004). For example, Fung (2003) validates technological proximity measured through an overlap of patent citations by showing that technological proximity is associated with stock return comovement.[8] Despite this evidence, a growing number of studies raise questions about the bias and imprecision of these technological proximity measures, ranging from sample selection concerns (Jaffe, Trajtenberg, and Forgarty 2000; Benner and Waldfogel 2008; Khoury and Bekkerman 2016) to the subjectivity of the patent class assignment owing to the influence of patent attorneys and patent examiners (Alcacer and Gittelman 2006; Alcacer, Gittelman, and Sampat 2009) and to firms' citing strategies (Lampe 2012). Lampe, for example, provides evidence that patent applicants strategically withhold between 21% and 33% of the relevant citations.

Another concern with the citation- and class-based measures is that they do not fully exploit the richness of patent disclosures. Our text-based measure of technological proximity reduces the noise and biases stemming from an inadequate patent taxonomy and the subjectivity of patent class assignments influenced by patent examiners and attorneys. In addition, because our measure uses the entirety of the information in patent descriptions, it detects innovation links that cannot be uncovered by traditional class- and citation-based measures.[9]

1.4 Textual analyses

Our paper is related to the growing body of research that employs textual analysis to uncover the role of public disclosures in financial markets. Prior

7 Extant research generally uses one of two approaches for measuring proximity. First, a technological domain is inferred from the 3-digit patent classification code that is listed on each patent and provided by USPTO (Jaffe 1986; Ahuja 2000; Song, Almeida, and Wu 2003; Bloom, Schankerman, and Van Reenen 2013); two firms are deemed to be close to each other if they tend to patent in the same technological domains. Second, researchers use citation overlap between firms to infer technological similarity (Stuart and Podolny 1996; Mowery, Oxley, and Silverman 1996).

8 However, Fung (2003) does not document return predictability.

9 Packalen and Bhattacharya (2012) and Younge and Kuhn (2016) use a similar text-based measure. Packalen and Bhattacharya use patent text to analyze the value of research efforts and the dynamics of the innovative research path. Younge and Kuhn develop a machine-automated measure of patent-to-patent similarity and show that the USPTO patent classification is too coarse for many of the applications sought in academic research.

studies have performed such analyses of disclosures filed with the SEC (Lehavy, Li, and Merkley 2011; Li 2008; Loughran and McDonald 2011, 2014; Biddle, Hilary, and Verdi 2009), conference calls (Larcker and Zakolyukina 2012), and business press (Tetlock, Saar-Tsechansky, and Macskassy 2008; Garcia 2013). For example, Hoberg and Phillips (2010, 2016) construct a measure of product market similarity and develop a new, dynamic TNIC by applying textual analysis to the product descriptions disclosed in 10-K filings. The authors show that their TNIC reflects product market similarity better than the traditional static SIC. We add to this body of work by using textual analysis of patents' Big Data to construct a novel measure of innovation similarity, which we show is superior to the traditional USPTO measures (Jaffe 1986; LSWZ 2019).[10]

2. Data, Methods, and Measure Validation

In this section, we describe the procedure used to estimate our Big Data–based measure of innovation similarity and report a number of tests aimed at validating this novel measure. To conserve space, the Internet Appendix chronicles our sample selection strategy and identifies our data sources.

2.1 Measuring innovation similarity among firms

To detect technologically similar firms, we leverage the latest advancements in Big Data science to automate the analysis of around 400 gigabytes of textual data containing descriptions from 7.7 million publicly available patents granted from 1976 to 2015 by the USPTO.[11] Our methods yield a unique continuous patent-to-patent innovation space in which topically similar patents are linked to each other. To describe a firm's innovation space, we aggregate individual patents at the firm level and use patent-to-patent similarity to construct a novel measure of technological distance among firms. Below we outline how we construct our new text-based measure of technological similarity, while the Internet Appendix details the technical steps.

To capture technological similarity, we represent each patent as a collection of professional terminology. We follow the data science literature (e.g., Gabrilovich and Markovitch 2009) and use external sources such as Wikipedia and professional dictionaries to establish the professional terminology for every patent in our sample. This process enables us to define two patents as similar if they share the same professional terminology.

To assess similarities of patent pairs, we use two MapReduce tasks (Elsayed, Lin, and Oard 2008). The first MapReduce task detects

[10] Outside financial economics, other papers that use textual analyses to establish patent or other types of links between firms include Gerken and Moehrle (2012); Moehrle and Gerken (2012); Park, Yoon, and Kim (2013); Zhanget al. (2016); Lin, Wu, and Hong (2016); and Arts, Cassiman, and Gomez (2018).

[11] The data are at https://figshare.com/s/d0319b79c316b034f72a

candidate professional terms in patents (the Map phase). We remove "boilerplates" (i.e., long lists of terminology used in patent texts to illustrate the invention generality). For each detected term, a list of patents that contain that term is constructed. The second MapReduce task runs through the list of patents produced by the Map phase and computes the distance (L_1) between every two patents. This distance is represented as a vector of their common terms weighted by a TFIDF (term frequency–inverse document frequency) variant (Manning and Schütze 1999). Terms that occur in the patent claims are assigned a greater weight than those that occur elsewhere in the patent text.

Our methodology overcomes the major barrier of constructing semantic similarity spaces for large-scale document repositories (i.e., the computational barrier). Our technique also reduces computation time because we assess only patent pairs that share specialized terminology and ignore most patent pairs that have no chance of being similar. As a result, our procedure outperforms many commonly accepted full-text comparison techniques which might work well in domains of short documents but throttle in the domain of patents—where documents are so long that the natural sparsity of texts does not hold.

We assess the patent text–based technological similarity between two firms from the patents they own. Firms A_1 and A_2 are considered similar if A_1 owns a patent n_1 that is similar to a patent n_2 owned by firm A_2. We detect the similarity between A_1 and A_2 starting from the year of publication of the newer of the two patents n_1 and n_2. If A_1 and A_2 own more than one pair of similar patents, the similarity between A_1 and A_2 is assessed starting with the oldest pair. Our continuous similarity measure for firms A_1 and A_2 is the log-weighted version of cosine similarity between their patents. Specifically, we take the log of the sum of similar patent pairs discounted by the age of the newer patent in each pair and normalize it by the log of the product of the total number of patents for each firm in the pair.[12] This measure considers the entire patent stock of a given firm to reflect its innovation history; however, it assigns a greater weight to newer patents to capture the greater relevance of more recent innovation activities.[13]

2.2 Sample description

Our main sample for the return predictability analyses includes firms with stock prices above $5 and available stock return, financial, and patent data

[12] We discount each pair by the factor, where is the age of the newer patent in the pair in a given year and T, the depreciation period, equals 20 years (the legal life of a patent). For example, if firm A_1 has a patent that was issued in 1995 and firm A_2 has a semantically similar patent that was issued in 1992, the discounting factor for this patent pair equals 1, 0.95, 0.90, and 0 for 1995, 1996, 1997, and 2015, respectively.

[13] Our results are not sensitive to the discounting factor. We obtain qualitatively similar results with $T = 5$, 7, and 10 years.

during 1977–2016.[14] We impose a threshold of $5 so that the stock prices will satisfy a minimum liquidity requirement, in order to ensure that our return predictability results are not driven by the thin market of illiquid securities.[15] To allow for further differentiation between previously documented industry momentum and technological cross-predictability, firms must have a TNIC classification (Hoberg and Phillips 2010, 2016). The TNIC classification provides a more precise measure of industry classification and is available only from 1996 to 2013. Therefore, we first examine the stock return predictability on the full 1977–2016 sample and then repeat the analysis with additional refinements as a robustness check for 1997–2014, the period for which product similarity controls are available.

Table 1, panel A reports descriptive statistics for the full sample during 1977–2016. The sample comprises, on average, 905.77 firms per year representing approximately 50% of the CRSP–COMPUSTAT common stock universe in terms of market capitalization. On average, each firm has 69 text-based, 270 class-based, and 93 citation-based technological peers. During 1997–2014, the period for which the Hoberg and Phillips classification is available, each firm has approximately 58 peers that operate in the same product space as defined by the TNIC classification. The text-based technological similarity has a mean of 0.12 and a standard deviation of 0.05, while the class-based technological similarity has a mean of 0.23 and a standard deviation of 0.09.[16]

Panel B provides descriptive statistics for the different industries that individually represent more than 1% of the firms in the 1977–2014 sample. Overall, these firms show a diverse industry representation. Not surprisingly, Electronic Equipment, Pharmaceutical, and Computer Software are the three most represented industries because of the prolific innovation activity in these areas.

2.3 Validation of the Big Data text-based similarity measure

We perform several tests designed to support the construct validity of our Big Data text-based innovation similarity measure.

2.3.1 Product similarity. We study the persistence of text-based technological similarity, substantiate the contention that technological and product similarity capture distinct economic links, and verify prior claims in the literature that innovation efforts precede product commercialization (Teece

[14] Our sample starts from July of 1977 because we match firms to CRSP stock returns from July year $t+1$ to June year $t+2$ to ensure that accounting and patent information is publicly available.

[15] The tenor of our results remains qualitatively the same without the $5 threshold.

[16] The average class-based connections per focus firm and the distribution of the class-based technological similarity measure are comparable with those reported by LSWZ (2019).

Table 1

Descriptive statistics for all NYSE/AMEX/NASDAQ firms with securities codes 10 and 11 contained in the CRSP/COMPUSTAT merged database

A. Firm characteristics: sample 1977–2016

Characteristic	Mean	STD	P10	P25	P50	P75	P90
# of firms per year	905.77	299.14	380	834	953	1117	1221
MV per year % of COMPUSTAT	0.51	0.05	0.44	0.50	0.52	0.55	0.55
N patents	661.06	2396.00	5	17	60	300	1413
N_links TS-text	69	88.96	5	17	60	300	1413
N_links TS-cite	93	120.81	12	19	36	79	158
N_links TS-class	270	159.39	12	25	51	109	213
N_links PS [(a)1]	58	63.87	97	165	242	343	475
Text-based similarity	0.12	0.05	0.07	0.09	0.11	0.14	0.16
Class-based similarity	0.23	0.09	0.14	0.16	0.20	0.27	0.37
Citation-based similarity	0.11	0.04	0.07	0.08	0.10	0.12	0.15
Product similarity[)1]	0.04	0.03	0.01	0.02	0.03	0.05	0.09
Log MV	7.20	2.04	4.59	5.74	7.12	8.56	9.90
BM	0.51	0.78	0.13	0.25	0.43	0.69	1.00
ROA	−0.01	0.30	−0.18	0.00	0.04	0.08	0.12
Complexity	0.05	0.05	0.03	0.03	0.04	0.06	0.09
Explorative	0.09	0.11	0.01	0.02	0.05	0.11	0.21
Exploitative	0.15	0.17	0.00	0.02	0.09	0.23	0.40

B. Descriptive statistics by industry: sample 1977–2016

Industry	*N obs. per year*	%	*Log MV*	*BM*	*ROA*	*N_links TS-text*	*N_links TS-class*	*N_links TS-cite*	*N_links PS*	*Complexity*	*Exploitative*	*Explorative*
Electronic Equipment	86.90	0.12	6.74	0.43	−0.02	82	261	111	87	0.04	0.12	0.09
Pharmaceutical	76.68	0.11	6.91	0.23	−0.19	64	252	61	77	0.12	0.30	0.09
Computer Software	55.56	0.08	7.05	0.30	−0.04	69	256	82	56	0.07	0.17	0.16
Machinery	51.56	0.07	6.70	0.44	0.04	49	229	92	52	0.03	0.10	0.04
Medical Equipment	36.97	0.05	6.37	0.32	−0.02	43	210	70	37	0.05	0.21	0.06
Computers	34.98	0.05	7.05	0.40	−0.02	100	288	141	35	0.05	0.15	0.08
Chemicals	34.70	0.05	7.47	0.39	0.05	94	318	125	35	0.05	0.17	0.06

(continued)

Table 1
Continued

B. Descriptive statistics by industry: sample 1977–2016

Industry	*N obs. per year*	%	*Log MV*	*BM*	*ROA*	*N_links TS-text*	*N_links TS-class*	*N_links TS-cite*	*N_links PS*	*Complexity*	*Exploitative*	*Explorative*
Measuring and Control equipment	28.90	0.04	6.37	0.41	0.01	58	301	92	29	0.05	0.13	0.07
Petroleum and Natural Gas	25.23	0.04	9.03	0.46	0.05	83	248	102	25	0.04	0.15	0.09
Automobiles	21.63	0.03	7.45	0.49	0.04	94	332	140	22	0.03	0.09	0.05
Business Service	21.36	0.03	6.36	0.32	−0.04	39	243	45	21	0.08	0.16	0.14
Consumer Goods	20.23	0.03	7.53	0.37	0.04	76	253	105	20	0.04	0.20	0.04
Business Supplies	17.95	0.02	7.50	0.44	0.04	64	308	110	18	0.03	0.11	0.06
Electrical Equipment	16.95	0.02	6.81	0.39	0.00	79	314	131	17	0.04	0.13	0.06
Construction Materials	16.49	0.02	6.86	0.45	0.05	45	265	77	16	0.03	0.09	0.08
Food Products	14.15	0.02	8.52	0.31	0.07	52	206	61	14	0.04	0.16	0.10
Communication	13.85	0.02	9.28	0.38	0.01	111	338	128	14	0.05	0.13	0.15
Steel Works	13.61	0.02	7.15	0.55	0.03	52	316	70	14	0.03	0.08	0.05

Stocks with a price less than \$5 at the end of the previous fiscal year are excluded. The sample consists of 290,572 firm-month observations for the 1977–2016 period. Panel A provides descriptive statistics for key variables. Panel B provides the mean of key variables sorted by the Fama-French 49 industries that each represents more than 1% of the firms in the sample. All variable definitions are in the Appendix. BM = book-to-market ratio; MV = market value; obs. = observations; PS = product market similarity; ROA = return on assets; TS-text = text-based technological similarity, TS-cite = citation-based technological similarity.

[1]PS is available only for 1997–2014.

Table 2
Validation of text-based technological similarity

A. Persistence of technological similarity: sample for 1997–2014 period

	(1) *Indicator PS* $_{t+1}$		(2) *Indicator TS-text* $_{t+1}$	
	Coefficient	*Odds ratio*	*Coefficient*	*Odds ratio*
Indicator PS_t	5.280***	196.43	−0.676***	0.51
	(175.754)		(−21.518)	
Indicator TS-text$_t$	0.088***	1.09	5.877***	356.60
	(4.507)		(208.473)	
Intercept	−4.852***	0.008	−4.569***	0.01
	(−178.961)		(−140.187)	
Obs. (N)	14,840,292		14,840,292	
Pseudo R^2	42.73%		68.29%	

B. Propensity of citation: sample for 1977–2016 period

	(1) *Indicator Cite* $_{t+1}$		(2) *Indicator Cite* $_{t+1}$		(3) *Indicator Cite* $_{t+1,t+3}$		(4) *Indicator Cite* $_{t+1,t+3}$	
	Coefficient	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*
Indicator TS-text$_t$	0.240***	1.27	2.046***	7.73	0.591***	1.81	1.911***	6.76
	(17.626)		(74.854)		(38.152)		(69.534)	
Indicator Cite$_t$	6.396***	599.55			5.523***	250.36		
	(224.210)				(205.110)			
Intercept	−4.857***	0.007	−5.105***	0.006	−3.880***	0.020	−4.061***	0.017
	(−156.206)		(−163.908)		(−124.951)		(−128.635)	
Obs. (N)	19,947,966		17,055,298		19,947,966		17,055,298	
Pseudo R^2	73.02%		4.67%		65.21%		4.34%	
	Full sample		Exclude pairs with common citations in year t		Full sample		Exclude pairs with common citations in year t	

C. Contemporaneous and predictive relation between fundamentals of the technologically similar firms: sample for 1977–2016 period

	(1) $ROA_{i,t}$	(2) $ROA_{i,t+1}$	(3) $RD_{i,t}$	(4) $RD_{i,t+1}$
$ROA_{i,t}^{TS-text}$	0.432***	0.217***		
	(5.172)	(3.531)		
$ROA_{i,t-1}$	0.540***	0.513***	0.024	0.031*
	(10.009)	(6.498)	(0.758)	(1.985)
$RD_{i,t}^{TS-text}$			0.298***	0.302***
			(4.118)	(2.831)
$RD_{i,t-1}$			0.748***	0.619***
			(29.463)	(6.395)
$BM_{i,t-1}$	0.008	−0.036***	−0.020***	−0.012
	(0.772)	(−3.799)	(−4.139)	(−1.616)
$LogMV_{i,t-1}$	0.023***	0.015***	−0.003***	−0.005***
	(8.260)	(5.671)	(−4.063)	(−2.792)
Industry and year fixed effects	Yes	Yes	Yes	Yes
Obs. (N)	24,044	23,132	24,047	24,080
R^2	43%	36.8%	82.2%	51.5%

D. Probability of forming acquirer-target pairs and technological similarity: sample of M&A activity in the period 1984–2018

	(1)		(2)		(3)		(4)		(5)	
	Coefficient	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*
Indicator TS-text	2.901***	18.187	3.005***	20.179	2.225***	9.255				
	(13.250)		(13.128)		(9.488)					
Same industry	3.629***	37.690	3.673***	39.380	3.300***	27.116	3.286***	26.729	3.693***	40.159
	(23.587)		(23.521)		(20.538)		(20.107)		(26.482)	
*Indicator TS-text*Same industry*			−0.629	0.533						
			(−1.311)							
Indicator TS-class					1.580***	4.855				
					(12.679)					
Indicator TS-text-NOT-class									0.945**	2.573
									(2.568)	

TS-text							11.996***	162000.0		
							(7.408)			
TS-class							3.664***	39.024		
							(9.871)			
Acquirer characteristics										
Size	0.496***	1.642	0.498***	1.645	0.452***	1.572	0.477***	1.612	0.538***	1.713
	(16.767)		(16.947)		(14.614)		(15.244)		(19.669)	
Leverage	−0.281	0.755	−0.284	0.753	−0.121	0.886	−0.367	0.693	−0.482	0.618
	(−0.924)		(−0.932)		(−0.385)		(−1.107)		(−1.592)	
ROA	1.966***	7.143	1.899***	6.678	1.464***	4.323	1.463***	4.321	1.962***	7.113
	(5.954)		(5.582)		(3.419)		(2.703)		(5.559)	
BM	−0.145*	0.865	−0.142*	0.868	−0.167**	0.846	−0.188**	0.828	−0.112***	0.894
	(−1.695)		(−1.919)		(−2.502)		(−2.382)		(−2.693)	
Cash	0.288	1.333	0.272	1.313	0.219	1.245	0.144	1.155	0.253	1.288
	(0.824)		(0.783)		(0.572)		(0.378)		(0.799)	
R&D intensity	0.522	1.686	0.540	1.717	−0.193	0.825	−0.612	0.542	1.209**	3.350
	(0.747)		(0.781)		(−0.240)		(−0.731)		(1.989)	
N patents	−0.000	1.000	−0.000	1.000	−0.000	1.000	−0.000	1.000	0.000	1.000
	(−1.402)		(−1.547)		(−1.324)		(−0.239)		(1.236)	
Target characteristics										
Size	−0.097***	0.907	−0.098***	0.906	−0.116***	0.891	−0.107***	0.899	−0.056**	0.946
	(−3.518)		(−3.533)		(−3.810)		(−3.631)		(−2.279)	
Leverage	−0.393	0.675	−0.382	0.682	−0.408	0.665	−0.309	0.734	−0.502**	0.606
	(−1.500)		(−1.458)		(−1.466)		(−1.125)		(−2.190)	
ROA	0.585**	1.795	0.578**	1.783	0.581*	1.789	0.678**	1.970	0.573**	1.774
	(2.012)		(1.998)		(1.794)		(2.144)		(2.054)	
BM	0.158**	1.171	0.161**	1.175	0.180**	1.197	0.174***	1.190	0.131**	1.140
	(2.486)		(2.526)		(2.559)		(2.608)		(2.416)	
Cash	−0.305	0.737	−0.299	0.742	−0.254	0.775	−0.273	0.761	−0.221	0.802
	(−1.053)		(−1.032)		(−0.795)		(−0.859)		(−0.876)	
R&D intensity	0.188	1.207	0.194	1.214	−0.046	0.955	0.020	1.020	0.306	1.359
	(0.360)		(0.373)		(−0.077)		(0.034)		(0.660)	
Patents (N)	−0.000***	1.000	−0.000***	1.000	−0.000***	1.000	−0.000***	1.000	−0.000***	1.000
	(−3.992)		(−4.080)		(−3.504)		(−3.981)		(−2.631)	
Obs. (*N*)	7268		7268		7268		7268		7268	

(continued)

Table 2
Continued

D. Probability of forming acquirer-target pairs and technological similarity: sample of M&A activity in the period 1984–2018

	(1)		(2)		(3)		(4)		(5)	
	Coefficient	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*	*Coefficient*	*Odds ratio*
Number of actual deals	1228		1228		1228		1228		1228	
Number of control deals	6040		6040		6040		6040		6040	
Pseudo R^2	0.621		0.621		0.661		0.658		0.561	

This table reports tests designed to provide support for the construct validity of the text-based technological similarity measure. Panel A displays the results of logistic regressions of the product and technological similarity in year *t+1* on product and technological similarity in year *t*. The sample consists of 14,840,292 firm-pairs and spans 1997 to 2014, the period for which the measure of product similarity is available. Panel B displays the results of logistic regressions of citations in year *t+1* (columns 1 and 2) and in the 2-year period from years *t+1* to *t+3* (columns 3 and 4) on the text-based technological similarity in year *t*. The logistic regressions in columns 1 and 3 use the sample of 19,947,966 observations, which includes all available firm-pairs in the years 1977–2016. The logistic regressions in columns 2 and 4 use the sample of 17,055,298 observations, which excludes firm-pairs with common citations in the previous years. Panel C shows the results of logistic regressions of the contemporaneous and predictive regressions of a focal firm's profitability and R&D intensity on the profitability and R&D intensity of its text-based technological peers. The panel regressions are estimated with the fixed effects for industry and year. Panel D analyzes whether text-based technological similarity predicts the probability of forming acquirer-target pairs. The sample consists of 1228 acquirer-target pairs with patents as well as a control group, which is formed by pairing every true target firm with five randomly drawn control firms for the acquirer and by pairing every true acquirer with five randomly drawn control firms for the target firm. The sample selection for M&A activity is described in Internet Appendix with the descriptive statistics presented in Table A3. The dependent variable equals one for the acquirer-target pair and zero for the control firm pairs. In column 1, the key independent variable is *Indicator TS-text*, which equals one if a pair is text-based technologically similar and zero otherwise. In column 2, we interact the *Indicator TS-text* with the *Same industry* indicator, the latter of which equals one if a pair belongs to the same three-digit SIC, and zero otherwise. In column 3, we add an additional explanatory variable *Indicator TS-class*, which equals one if a pair of firms is class-based technologically similar and zero otherwise. In column 4, the key explanatory variable is *Indicator TS-text-NOT-class*, which equals one if a pair of firms is text-based technologically similar but not class-based technologically similar and zero otherwise. In column 5, the key explanatory variables are continuous measures of text- and class-based technological similarity, *TS-text* and *TS-class*, respectively. The vector of control variables includes the indicator variable *Same industry* and various financial performance characteristics of the acquirer and target firms. All variables are defined in the Appendix. In panels A–C, *t*-statistics are reported in parentheses and are based on standard errors that are double-clustered by firm and year (Thompson 2011; Petersen 2009), while in panel D the standard errors are clustered by M&A deal bids.
***The coefficient estimates are different from zero at 1% level.

1996; Alexopoulos 2011). Panel A of Table 2 reports the results of two logistic regressions that analyze the sample of 14,840,292 firm-pairs running from 1997 to 2014, the period for which the measure of product similarity is available. In column 1, the dependent variable is an *Indicator* PS_{t+1}, which is set to one if a given pair of firms belongs to the same TNIC industry in year $t+1$ and set to zero otherwise. In column 2, the dependent variable is an *Indicator* $TS\text{-}text_{t+1}$, which is set to one if a given pair of firms are text-based technologically similar in year $t+1$. The independent variables are *Indicator* PS_t and *Indicator* $TS\text{-}text_t$. We find that both technological and product similarity are highly persistent. For example, in column 1, the estimate on the indicator of product similarity in year t, *Indicator* PS_t, equals 5.280, while in column 2, the estimate on *Indicator* $TS\text{-}text_t$, equals 5.877.

We also find that technological similarity predicts product similarity: technologically similar firms in year t are more likely to become product market neighbors in the following year, as evidenced by the positive and statistically significant coefficient of 0.088 on *Indicator* $TS\text{-}text_t$ in column 1. In contrast, firms that belong to the same product space in year t are less likely to become technologically similar in the following year, as shown by the negative estimate −0.676 on the dummy variable for technological similarity in year t reported in column 2. These results suggest that the technology development phase precedes product commercialization: product market neighbors are less likely to become technology neighbors because they already operate in the same technology space.

2.3.2 Patent citations. We verify that text-based technologically similar firms are indeed peers by analyzing whether they are more likely to cite each other in the future. Panel B reports the logistic regression results where the dependent variables are *Indicator* $Cite_{t+1}$ (in columns 1 and 2) and *Indicator* $Cite_{t+1,\ t+3}$ (in columns 3 and 4). *Indicator* $Cite_{t+1}$ and *Indicator* $Cite_{t+1,\ t+3}$ are dummy variables that equal one if a pair of firms cite each other in year $t+1$ or during the 3-year period $t+1$ to $t+3$, respectively. Controlling for the citations in the previous years, we find that text-based innovation similarity predicts future citations by text-linked firms as evidenced by a positive and statistically significant coefficient estimate on *Indicator* $TS\text{-}text_t$ in columns 1 and 3, 0.240 and 0.591, respectively. Importantly, when we exclude firm-pairs with common citations in previous years, text-based technological similarity still predicts future citation by the text-linked firms as shown by the positive coefficient on *Indicator* $TS\text{-}text_t$ in columns 2 and 4. These effects are both statistically and economically significant. For example, the odds ratio of 7.73 in column 2 suggests that the odds of a given firm citing the focal firm's patents are 673% higher if that firm is text-based technologically linked with the focal firm.

2.3.3 Economic connections. We show the existence of real economic connections between technologically similar firms, which are an important precondition for asset pricing implications. Specifically, we document (i) the contemporaneous correlation and predictive relation of the profitability of technologically linked firms and (ii) the contemporaneous and predictive relation of innovation-related activity along technological connections. We measure the profitability of a focal firm *i* with its return on assets, $ROA_{i,t}$, which is computed as the earnings before interest, taxes, depreciation, and amortization scaled by total assets. For each focal firm *i* we calculate ROA of its text-based technologically similar peers, $ROA_{i,t}^{TS-text}$, as the average ROA of all firms that are technologically linked with firm *i*, weighted by the strength of their technological similarity.

The regression results are reported in panel C of Table 2. In column 1, the dependent variable is focal firm *i*'s profitability in year *t*, $ROA_{i,t}$, while in column 2, the dependent variable is firm *i*'s ROA in year t+1. The main independent variable is the contemporaneous ROA of its technologically linked peers, $ROA_{i,t}^{TS-text}$. Additionally, we control for firm *i*'s own lagged profitability, $ROA_{i,t-1}$, its size, $LogMV_{i,t-1}$, and its book-to-market ratio, $LogBM_{i,t-1}$. To control for the economic cycle and industry profitability, we include industry and year fixed effects. The standard errors are double clustered by firm and year (Thompson 2011; Petersen 2009). The contemporaneous correlation between ROA of technologically similar firms is confirmed by the positive and statistically significant coefficient on $ROA_{i,t}^{TS-text}$ (0.432) in column 1, while the predictive relation is supported by the positive coefficient on $ROA_{i,t}^{TS-text}$ (0.217) in column 2.

We use R&D intensity (*RD*), computed as the ratio of R&D expenditures to total assets at the end of the fiscal year, as a proxy for a firm's innovation activity. Columns 3 and 4 of panel C present the results when the dependent variable is the focal firm *i*'s R&D intensity (*RD*) in the years *t* and t+1, respectively. For each focal firm *i*, we calculate the R&D intensity of its technologically linked peers, $RD_{i,t}^{TS-text}$, as the average *RD* of all firms that are technologically connected with firm *i*, weighted by the strength of their technological similarity. Confirming the existence of real effects of technological connections on a firm's innovation activity, we find that the R&D intensity of technologically linked firms is contemporaneously correlated with (and predicted by) the prior year R&D intensity of their innovation peers.

2.3.4 Probability of forming M&A pairs. We examine the effect of innovation similarity on the probability of M&A bids. This validation test is based on the rapidly growing literature on the effect of innovation on M&A activity (Phillips and Zhdanov 2013; Sevilir and Xuan 2012). Bena and Li (2014), for example, show that firms are more likely to merge if they exhibit technological overlap, as proxied by the closeness of their patent portfolios, their shared knowledge bases, or their reciprocal patent citations. Based on these findings, they conclude that the synergies attainable from the combination of innovation capabilities promote merger activity. Building on this idea we validate our novel measure by showing that text-based technological similarity predicts the probability of forming M&A pairs beyond technological overlap captured by patent classes.

The sample selection for M&A activity is described in the Internet Appendix. Because technological similarity is only possible between innovative firms, we perform these tests on the sample of acquirer/target pairs with innovation activities captured by patent filing (1,228 firm-pairs). We follow the approach in Bena and Li (2014) to create a control sample by pairing each true target firm with five randomly drawn control firms for the acquirer and by pairing each true acquirer with five randomly drawn control firms for the target firm. All control firms are drawn from the population of CRSP/COMPUSTAT observations that (i) have data to compute the required variables for the analysis, (ii) are neither an acquirer nor a target in the 3-year period prior to the bid, and (iii) are active in patenting.

Panel D reports the coefficient estimates for five conditional logistic regressions in which the dependent variable equals one for the acquirer-target firm-pair and to zero for the control firm-pairs. The key independent variable of interest in column 1 is *Indicator TS-text*, a dummy that is set to one if a pair is text-based technologically similar and to zero otherwise. We find a positive and significant association between text-based similarity and the likelihood of forming an M&A pair, as evidenced by the positive coefficient estimate on *Indicator TS-text* (2.901, z-statistic = 13.250). This result confirms the findings of Bena and Li (2014) that firms are more likely to merge if they exhibit technological overlap derived from either patent classification or reciprocal patent citations. The increased likelihood of text-based innovation peers forming M&A pairs also provides additional validation for our novel Big Data text-based measure of innovation similarity.

In column 2 of panel D, we study the join effect of product market similarity and innovation similarity on the probability of M&A bids. We find that product market relatedness is an important driver of M&A decisions as evidenced by the positive and statistically significant coefficient on *Same Industry* (3.673, z-statistic = 23.521). Yet the insignificant coefficient on the interaction term *Indicator TS-text**Same Industry* indicates that overlap in two core activities—product and innovation—has no material impact on M&A decisions. This result contrasts with those in Bena and Li (2014) showing that the positive impact of technological (or product) similarity is reduced, but not fully eliminated, for pairs that overlap in both.[17]

[17] We attribute this discrepancy in the findings to the fact that our procedures to identify both product similarity and technological similarity differ from those in Bena and Li (2014).

Column 3 augments our specification with *Indicator TS-class*, a dummy variable that equals one if a pair is class-based technologically similar and zero otherwise. The estimates show that while the traditional class-based measure of technological similarity is an important predictor of the likelihood of acquirer/target pair formation, it does not diminish the effect of the text-based innovation similarity measure. Specifically, the coefficient on *Indicator TS-text* (2.225) is close to the estimate for the same variable in column 1 (2.901). In column 4, we repeat our analysis using continuous measures of text- and class-based technological similarity. The strength of text-based technological similarity remains positively related to the likelihood of an M&A bid.

In column 5, we study "pure" text-based technologically similar target and acquirer firms. For this purpose, we remove the overlap between text- and class-based technologically similar firms and evaluate the likelihood of M&A pair formation on the *Indicator TS-text-NOT-class*, which equals one if a pair is text-based technologically similar but is not class-based similar. The estimate on *Indicator TS-test-NOT-class* (0.945, t-statistic = 2.568) indicates that firms linked by our text-based innovation similarity measure (that do not overlap with the class-based method) are more likely to merge than firms that are not linked. This result is not only statistically significant but economically important. The estimated odds ratio of 2.573 indicates that the odds of a given pair being an acquirer/target pair are 157% percent higher if they are "pure" text-based technologically similar firms.

In sum, our analysis of the likelihood of forming M&A pairs validates our novel Big Data text-based measure and strongly indicates that the measure provides additional valuable information that helps predict M&A deals. Notably, our Big Data text-based method identifies target/acquirer pairs that account for US$360 billion in terms of M&A deal value that the traditional patent-class-based method misses.

3. Baseline Empirical Analyses

3.1 Contemporaneous comovement of technologically similar firms

If investors incorporate publicly available information about innovation links in stock prices, then a focal firm's stock prices should adjust to reflect information about its technologically similar peers. As a result, stock prices of technologically similar firms should comove contemporaneously.

In Table 3, we estimate six different Fama and MacBeth (1973) regressions of focus firm i's stock return (in percentage) in month t, $r_{i,t}$, on the portfolio return of firm i's technologically similar peers in month t, $r_{i,t}^{TS-text}$, calculated as the average return of firm i's text-based technologically linked peers, weighted by the strength of their similarity.

To ensure that our results are not induced by the comovement of stock returns in firm i's product market space, we control for $r_{i,t}^{Ind}$, which is the

Table 3
Contemporaneous comovement of stock returns

	(1) $r_{i,t}$	(2) $r_{i,t}$	(3) $r_{i,t}$	(4) $r_{i,t}$	(5) $r_{i,t}$	(6) $r_{i,t}$
$r_{i,t}^{TS-text}$	2.932***			1.959***	2.127***	1.508***
	(20.899)			(15.249)	(14.947)	(10.541)
$r_{i,t}^{TS-cite}$		2.585***			1.654***	1.179***
		(20.020)			(13.204)	(10.179)
$r_{i,t}^{TS-class}$			3.425***	2.323***		1.996***
			(8.922)	(6.245)		(4.996)
$r_{i,t}^{IND}$	2.026***	2.064***	1.984***	1.839***	1.906***	1.770***
	(38.679)	(38.490)	(40.518)	(38.545)	(37.690)	(37.408)
$r_{i,t-1}$	−0.817***	−0.748***	−0.800***	−0.825***	−0.778***	−0.799***
	(−7.799)	(−7.523)	(−7.877)	(−8.031)	(−7.700)	(−7.954)
$r_{i,t-12,t-2}$	0.201*	0.216*	0.202*	0.183*	0.202*	0.171
	(1.795)	(1.888)	(1.823)	(1.685)	(1.775)	(1.557)
BM	0.138	0.123	0.182	0.186	0.131	0.155
	(1.124)	(1.032)	(1.602)	(1.608)	(1.104)	(1.397)
$LogMV$	−0.072	−0.075	−0.068	−0.067	−0.074	−0.070
	(−1.468)	(−1.495)	(−1.456)	(−1.411)	(−1.478)	(−1.462)
Constant	1.669***	1.623***	1.406***	1.376***	1.631***	1.333***
	(3.635)	(3.568)	(3.557)	(3.385)	(3.700)	(3.228)
Obs. (N)	291,749	284,704	285,506	285,506	284,704	284,071
Adj. R^2	0.116	0.117	0.118	0.126	0.125	0.133

This table presents the results of the time-series averages of the coefficient estimates from monthly cross-sectional contemporaneous regression of a focal firm's stock returns ($r_{i,t}$), in percentages, on the portfolio returns of its text-based technological peers ($r_{i,t}^{TS-text}$), citation-based technological peers ($r_{i,t}^{TS-cite}$), class-based technological peers ($r_{i,t}^{TS-class}$), industry portfolio returns ($r_{i,t}^{IND}$), the focal firm's stock return in the previous month ($r_{i,t-1}$), the focal firm's stock returns over the preceding 11 months ($r_{i,t-12,t-2}$), a natural logarithm of the focal firm's market value ($LogMV_{i,t}$), and the focal firm's book-to-market ratio ($BM_{i,t}$). All variable definitions are in the Appendix. All returns are in excess of the risk-free rate and are standardized to have standard deviation of one. The regressions use the main sample, which spans 1977–2016. t-statistics are reported in parentheses, and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation.
***The coefficient estimates are different from zero at 1% level.

value-weighted Fama-French (1997) 49-industry portfolio return. In addition, we control for the traditional measures of technological proximity (citation overlap and USPTO class-cosine similarity) by including $r_{i,t}^{TS-cite}$ and $r_{i,t}^{TS-class}$. The first variable is the average portfolio return of firms that have citation overlaps with firm i, weighted by the strength of the citation overlap. The second variable is the average portfolio return of firms that have a greater than zero class-cosine similarity to firm i, weighted by the strength of the class-cosine similarity. Other independent variables control for the short-term reversal and medium-term continuation of the stock return (Jegadeesh and Titman 1993) and for a firm's cross-sectional characteristics such as size and book-to-market ratio (Fama and French 1993). We use $r_{i,t-1}$, the return of firm i in the previous month, to control for short-term reversal and $r_{i,t-12,t-2}$ the average return of firm i over the preceding 11 months starting with month $t-12$, to control for medium-term continuation of the return.[18] To facilitate the comparison and interpretation of the magnitude of these coefficient estimates, we normalize all independent variables

18 All returns are monthly to maintain comparability across the stock return variables.

representing stock returns to have standard deviations equal to one. Standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation.

Table 3 reports the time-series average of each regression coefficient from monthly cross-sectional regressions and time-series *t*-statistics. The results show that a firm's stock return contemporaneously comoves with the portfolio of its text-based technologically similar peers, as evidenced by positive and statistically significant coefficient estimates on $r_{i,t}^{TS-text}$. For example, in column 6, which includes all control variables, the estimate on $r_{i,t}^{TS-text}$ is equal to 1.508. Coefficient estimates on the portfolios of stocks that are in the same industry and citation- and class-based technologically similar are also positive and statistically significant (1.770, 1.179, and 1.996, respectively). Importantly, the economic significance of the contemporaneous correlation between a focal firm and its text-based technologically linked portfolios is comparable to that between a focal firm and its industry portfolio. Contemporaneous comovement of technologically similar firms indicates that investors do indeed incorporate information about innovation similarity into stock prices. It also serves as additional confirmation of the construct validity of our novel text-based measure by showing that text-based technological similarity captures important economic links beyond within-industry relations.

3.2 Return cross-predictability—portfolio tests

To examine whether information about technological similarity is fully incorporated into stock prices we report the portfolio returns and profitability of self-financing trading strategies that are based on the lagged stock returns of technologically similar peers. We construct self-financing trading strategies as follows. At the beginning of each month, we sort stocks into deciles based on the previous month's returns on the portfolios of technologically similar firms. Next, for each decile, we form value- (equal-) weighted portfolios and calculate returns on these portfolios for the following month. Self-financing strategies consist of buying firms that earned high returns on technologically linked firms (top tenth decile) during the previous month and selling firms that earned low returns on linked firms (bottom first decile) during the previous month.

Table 4, panel A reports equal- (value-) weighted portfolio returns that provide strong evidence in support of return cross-predictability for innovation peers. The average monthly portfolio returns (net of a 1-month Treasury bill rate) increase monotonically with the prior month returns of technologically similar firms. Specifically, the mean excess return for the high equal-weighted portfolio is 1.60%, while the mean excess return for the low portfolio is 0.31%. A self-financing strategy of buying high portfolios and

Table 4
Stock return cross-predictability—portfolio analysis

A: Portfolio returns, sample 1977–2016

Decile	Excess Return (%)	CAPM α (%)	3-Factor α (%)	4-Factor α (%)	5-Factor α (%)	6-Factor α (%)
Low (1)	0.31	−0.49	−0.52	−0.25	−0.39	−0.22
	(0.99)	(−2.75)	(−3.12)	(−1.57)	(−2.29)	(−1.35)
2	0.51	−0.24	−0.30	−0.11	−0.24	−0.12
	(1.8)	(−1.77)	(−2.32)	(−0.89)	(−1.78)	(−0.9)
3	0.63	−0.09	−0.16	0.02	−0.06	0.05
	(2.36)	(−0.7)	(−1.34)	(0.16)	(−0.51)	(0.46)
4	0.79	0.10	0.01	0.20	0.03	0.16
	(3.07)	(0.78)	(0.08)	(1.83)	(0.24)	(1.49)
5	0.74	0.06	−0.05	0.09	−0.05	0.06
	(2.94)	(0.51)	(−0.49)	(0.9)	(−0.43)	(0.57)
6	1.09	0.41	0.33	0.44	0.33	0.42
	(4.29)	(3.34)	(3.11)	(4.14)	(3.11)	(3.97)
7	1.19	0.51	0.42	0.56	0.42	0.53
	(4.7)	(4.19)	(4.06)	(5.42)	(4.04)	(5.16)
8	1.19	0.53	0.46	0.57	0.49	0.57
	(4.48)	(4.32)	(4.31)	(5.38)	(4.59)	(5.44)
9	1.35	0.60	0.56	0.73	0.69	0.79
	(4.63)	(3.75)	(4)	(5.17)	(5)	(5.87)
High (10)	1.60	0.85	0.86	0.98	1.05	1.13
	(5.01)	(4.12)	(4.77)	(5.35)	(6.17)	(6.56)
H − L (equal-weighted)	**1.29**	**1.34**	**1.37**	**1.23**	**1.44**	**1.35**
	(5.03)	(5.17)	(5.3)	(4.63)	(5.46)	(5.06)
H − L (value-weighted)	**0.88**	**0.93**	**0.99**	**0.91**	**1.00**	**0.95**
	(3.13)	(3.25)	(3.46)	(3.08)	(3.36)	(3.13)

B: Risk-factor loading, sample 1977–2016

Decile	Alpha	MKT	SMB	HML	MOM
Low (1)	−0.25	1.13	0.53	−0.19	−0.31
	(−1.57)	(30.69)	(9.85)	(−3.35)	(−8.88)
High (10)	0.98	1.01	0.76	−0.29	−0.14
	(5.35)	(23.88)	(12.24)	(−4.35)	(−3.34)
H − L (equal-weighted)	**1.23**	**−0.12**	**0.23**	−0.09	**0.18**
	(4.63)	(−1.88)	(2.55)	(−0.99)	(3.01)
H − L (value-weighted)	**0.91**	−0.09	0.00	−0.15	**0.13**
	(3.08)	(−1.34)	(0)	(−1.46)	(1.95)

This table reports abnormal returns and factor loadings of the text-based technological similarity strategy. Panel A presents calendar-time portfolio abnormal returns. Firms are sorted into deciles at the beginning of each month based on returns of text-based technologically similar firms in the previous month ($r_{i,t}^{TS-text}$). Key variables are defined in the Appendix. Excess return is the difference between portfolio returns and the 1-month Treasury bill rate. Alpha is the intercept in a regression of monthly excess return on factor returns from the rolling strategy. Factor models are (i) CAPM, (ii) Fama-French three-factor model, which includes MKT (the market factor), SMB (the size factor), HML (the value factor), (iii) four-factor model (Fama-French three-factors and MOM [the momentum] factor), (iv) Fama-French five-factor model (Fama-French three-factors and RMW [robust minus weak] and CMA [conservative minus aggressive] factors), (v) Fama-French six-factor model (Fama-French five-factor plus MOM). Returns and alphas are reported in (monthly) percentages. Panel B reports the alpha and risk-factor loadings, where the benchmark is a four-factor model. *t*-statistics are reported in parentheses and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. Bold text indicates that the values are different from zero with at least 10% significance level.

shorting low equal- (value-) weighted portfolios yields a 1.29% (0.88%) monthly return, or about 17% (11%) in annualized returns.

To address concerns that the abnormal returns of self-financing strategies might be attributable to a systematic risk, we regress the excess portfolio return on the excess market return and on the well-known risk factors. In panel A, we report alphas for the capital asset pricing model (CAPM); the three-factor Fama-French model, which includes the excess market return, small-cap performance over large-cap performance (SML), and high-minus-low (HML) factors (Fama and French 1993); the four-factor model, which includes the previous three factors and the momentum factor (MOM; Carhart 1997); the five-factor Fama-French model, which augments the three-factor model with RMW (robust minus weak) and CMA (conservative minus aggressive) factors (Fama and French 2015); and the six-factor model, which adds MOM to the five Fama-French factors. We find that even risk-adjusted returns for portfolios formed based on the past month's returns of technologically connected peers increase monotonically. The hedge portfolio alphas appear to be unaffected by the risk adjustments. For example, the monthly alphas from the CAPM, three-, four-, five-, and six-factor models are 1.34%, 1.37%, 1.23%, 1.44%, and 1.35% respectively.

Panel B reports the hedge portfolio alphas and the risk-factor loading of the four-factor model. The hedge portfolio has a negative loading on the market return and positive loadings on both SMB and MOM. However, even after controlling for these risk exposures, the strategy still produces significant alphas. Overall, these results suggest that the trading strategy of buying (selling) stocks with high (low) lagged returns of their technological peers yields substantial subsequent returns after controlling for common risk factors.

3.3 Using various control variables and different subsamples

We now examine the return predictability by technological peers in a regression setting that allows us to control for various components identified by prior literature as predictive factors of a firm's stock returns. Table 5 reports Fama and MacBeth regressions where the dependent variable is the stock return (in percentage) of focus firm i in month $t+1$, $r_{i,t}$, and the independent variables are the same as in the contemporaneous comovement tests described in Section 3.1.

Panel A reports the results for the full 1977–2016 sample, while panel B displays the results for the 1997–2014 sample, where product similarity based on the TNIC classification is available. The results in both panels indicate that lagged returns of technologically similar peers cross-predict a focal firm's stock returns as shown by the positive and significant coefficient on $r_{i,t-1}^{TS-text}$ in all regression specifications. For example, the coefficient estimate on $r_{i,t-1}^{TS-text}$ is 0.916 in column 1, panel A, where we control for industry momentum, a

Table 5

Stock return cross-predictability—Fama-MacBeth regression analysis

A: Sample period 1977–2016

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$	(6) $r_{i,t+1}$
$r_{i,t}^{TS-text}$	0.916***			0.895***[1]	0.831***	0.837***
	(8.162)			(8.063)	(7.228)	(7.145)
$r_{i,t}^{TS-class}$		0.735***		0.303[1]		0.194
		(4.106)		(1.603)		(0.856)
$r_{i,t}^{TS-cite}$			0.585***		0.220**	0.183*
			(5.549)		(2.039)	(1.687)
$r_{i,t}^{IND}$	0.124**	0.183***	0.195***	0.116**	0.133**	0.130**
	(2.321)	(3.417)	(3.457)	(2.281)	(2.451)	(2.486)
$r_{i,t}$	−0.172*	−0.172*	−0.199*	−0.179*	−0.199*	−0.208*
	(−1.760)	(−1.790)	(−1.766)	(−1.864)	(−1.785)	(−1.910)
$r_{i,t-12,t-1}$	0.235**	0.236**	0.252**	0.231**	0.250**	0.236**
	(2.117)	(2.140)	(2.148)	(2.118)	(2.128)	(2.052)
LogBM	0.091	0.110	0.075	0.126	0.100	0.126
	(0.737)	(0.899)	(0.611)	(1.044)	(0.838)	(1.062)
LogMV	−0.081*	−0.084*	−0.081	−0.080*	−0.075	−0.074
	(−1.670)	(−1.737)	(−1.630)	(−1.653)	(−1.525)	(−1.510)
Constant	1.754***	1.779***	1.798***	1.735***	1.722***	1.607***
	(3.413)	(3.441)	(3.328)	(3.372)	(3.258)	(3.070)
Obs. (N)	290572	284351	283552	284351	283552	282921
Adj. R^2	0.082	0.084	0.084	0.089	0.089	0.096

B: Sample period 1997–2014

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$	(6) $r_{i,t+1}$	(7) $r_{i,t+1}$	(8) $r_{i,t+1}$	(9) $r_{i,t+1}$	(10) $r_{i,t+1}$	(11) $r_{i,t+1}$	(12) $r_{i,t+1}$
$r_{i,t}^{TS-text}$	0.969***	1.034***					1.103***[2]	1.208***[3]	1.040***	1.153***	1.135***	1.280***
	(5.807)	(7.234)					(8.649)	(9.996)	(6.872)	(8.963)	(8.889)	(11.356)
$r_{i,t}^{TS-class}$			0.344	0.174			−0.210[2]	−0.367*[3]			−0.191	−0.336
			(1.339)	(0.770)			(−0.868)	(−1.685)			(−0.766)	(−1.454)

(continued)

Table 5

Continued

B: Sample period 1997–2014

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$	$r_{i,t+1}$
$r_{i,t}^{TS-cite}$					0.437***	0.333**			−0.068	−0.175	−0.048	−0.138
					(3.074)	(2.571)			(−0.600)	(−1.511)	(−0.429)	(−1.140)
$r_{i,t}^{PS}$		0.207*		0.358***		0.348***		0.245**		0.227*		0.261***
		(1.744)		(3.612)		(2.841)		(2.569)		(1.959)		(2.784)
$r_{i,t}^{IND}$	0.096	0.051	0.186**	0.133*	0.171**	0.113	0.103	0.066	0.094	0.055	0.108	0.076
	(1.269)	(0.652)	(2.511)	(1.738)	(2.186)	(1.417)	(1.484)	(0.900)	(1.283)	(0.705)	(1.590)	(1.041)
$r_{i,t}$	−0.214	−0.208	−0.213	−0.210	−0.214	−0.214	−0.222	−0.216	−0.224	−0.224	−0.231	−0.228
	(−1.480)	(−1.335)	(−1.454)	(−1.321)	(−1.428)	(−1.331)	(−1.520)	(−1.372)	(−1.507)	(−1.403)	(−1.580)	(−1.444)
$r_{i,t-12,t-1}$	−0.014	−0.054	0.000	−0.047	−0.011	−0.059	−0.009	−0.055	−0.021	−0.067	−0.026	−0.074
	(−0.083)	(−0.315)	(0.002)	(−0.282)	(−0.063)	(−0.332)	(−0.059)	(−0.330)	(−0.121)	(−0.382)	(−0.156)	(−0.428)
BM	−0.076	−0.046	−0.045	−0.022	−0.054	−0.023	−0.023	−0.001	−0.030	−0.004	0.003	0.025
	(−0.469)	(−0.279)	(−0.272)	(−0.131)	(−0.316)	(−0.138)	(−0.140)	(−0.009)	(−0.180)	(−0.023)	(0.020)	(0.145)
LogMV	−0.148*	−0.145*	−0.148*	−0.145*	−0.151*	−0.145*	−0.146*	−0.143*	−0.147*	−0.143*	−0.144*	−0.141*
	(−1.843)	(−1.728)	(−1.835)	(−1.729)	(−1.840)	(−1.727)	(−1.813)	(−1.695)	(−1.818)	(−1.707)	(−1.781)	(−1.669)
Constant	2.094**	1.939**	1.955**	1.763**	2.174**	2.005**	1.922**	1.746*	2.088**	1.948**	1.927**	1.765*
	(2.422)	(2.196)	(2.260)	(1.992)	(2.412)	(2.220)	(2.217)	(1.958)	(2.384)	(2.188)	(2.205)	(1.965)
Obs. (*N*)	202569	168620	197884	164951	197698	164813	197884	164951	197698	164813	197145	164296
Adj. *R2*	0.059	0.066	0.061	0.067	0.059	0.066	0.064	0.070	0.062	0.069	0.066	0.073

This table reports the results of Fama-MacBeth return forecasting regressions. Panel A uses the 1977–2016 sample. The dependent variable is the focal firm's monthly return in percentages ($r_{i,t+1}$). The explanatory variables are portfolio returns of its text-based technological peers ($r_{i,t}^{TS-text}$), citation-based technological peers ($r_{i,t}^{TS-cite}$), and class-based technological peers ($r_{i,t}^{TS-class}$). Control variables are industry portfolio returns ($r_{i,t}^{IND}$), the focal firm's stock return in the previous month ($r_{i,t-1}$), the focal firm's stock returns over the preceding 11 months ($r_{i,t-12,t-2}$), a natural logarithm of the focal firm's market value ($LogMV_{i,t}$), and the focal firm's book-to-market ratio ($BM_{i,t}$). In columns 1, 2, and 3, the focal firm's monthly return ($r_{i,t+1}$) is regressed on control variables and $r_{i,t}^{TS-text}$, $r_{i,t}^{TS-class}$, and $r_{i,t}^{TS-cite}$. In columns 4 and 5, the focal firm's monthly return ($r_{i,t+1}$) is regressed on control variables and $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-class}$, $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-cite}$. In column 6, we regress the focal firm's monthly return ($r_{i,t+1}$) on $r_{i,t}^{TS-text}$, $r_{i,t}^{TS-class}$, $r_{i,t}^{TS-cite}$ and control variables. Panel B reports the results of Fama-MacBeth return forecasting regressions in which dependent variables are augmented by the portfolio returns of the text-based product-similar firms ($r_{i,t}^{PS}$). In columns 1 and 2, the explanatory variables are $r_{i,t}^{TS-text}$, and $r_{i,t}^{TS-text}$ and $r_{i,t}^{PS}$. In columns 3 and 4, the explanatory variables are $r_{i,t}^{TS-class}$, and $r_{i,t}^{TS-class}$ and $r_{i,t}^{PS}$. In columns 5 and 6, the explanatory variables are $r_{i,t}^{TS-cite}$, and $r_{i,t}^{TS-cite}$ and $r_{i,t}^{PS}$. In columns 7 and 8, the explanatory variables are $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-class}$ and $r_{i,t}^{TS-text}$, $r_{i,t}^{TS-class}$ and $r_{i,t}^{PS}$. In columns 9 and 10, the explanatory variables are $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-cite}$ and $r_{i,t}^{TS-text}$, $r_{i,t}^{TS-cite}$ and $r_{i,t}^{PS}$. In column 11, the explanatory variables are $r_{i,t}^{TS-text}$, $r_{i,t}^{TS-class}$ and $r_{i,t}^{PS}$. In all columns we include control variables. These regressions are estimated on the 1997–2014 sample, the period for which the measure of text-based product similarity is available. All variable definitions are in the Appendix. All returns are in excess of the risk-free rate and are standardized to have standard deviation of one. *t*-statistics are reported in parentheses and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. The symbols ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

[1]F-test of the equality coefficient estimates on $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-class}$: z-value = 7.28, p-value = .0072

[2]F-test of the equality coefficient estimates on $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-class}$: z-value = 23.11, p-value < .0001

[3]F-test of the equality coefficient estimates on $r_{i,t}^{TS-text}$ and $r_{i,t}^{TS-class}$: z-value = 39.98, p-value < .0001

firm's short-term reversal and medium-term continuation, and a firm's size and book-to-market ratio. Importantly, this estimate is also economically significant: it can be interpreted as a 91.6 basis points monthly premium on the self-financing trading strategy of buying (selling) stocks that earned high (low) returns on their technologically linked peers during the previous month. This finding is consistent with the yield of 129 (88) basis points on the equal- (value-) weighted hedge portfolio of self-financing trading strategies reported in Section 3.2.

Columns 2 and 3 in panel A of Table 5 report return predictability by technologically linked firms determined by the traditional class- and citation-based measures, respectively. While the coefficient estimates on the class- and citation-based variables are also statistically significant (0.735 and 0.585, respectively), their magnitude substantially weakens (0.303 and 0.220, respectively) when text-based technological similarity is used as a predictive factor in columns 4 and 5. We find that predictive ability of both traditional measures is lower than the text-based similarity. Moreover, class-based technological similarity is completely subsumed by text-based similarity as the insignificant coefficients on class-based portfolios in column 4 indicate. In contrast, the coefficient on $r_{i,t}^{TS-text}$ decreases only slightly (0.837) when we control for the industry momentum and for both traditional measures of technological similarity in column 6.

To ensure that our results are not due to the simple rediscovery of previously documented industry momentum, we control for the more rigorous measure of product market similarity of Hoberg and Phillips (2010) and report the results in panel B of Table 5. We confirm that the product similarity portfolios predict stocks returns of a focus firm as evidenced by positive and significant estimates on $r_{i,t-1}^{PS}$ in all tests. Notably, the control for product market similarity does not impact the predictive ability of text-based innovation portfolios. For example, for the 1997–2014 sample, without the product similarity control, the coefficient estimate on $r_{i,t}^{TS-text}$ is 0.969 (column 1), while in the product similarity portfolio it is 1.034 (column 2). Conversely, the coefficient estimates on class-based technological portfolios $r_{i,t}^{TS-class}$ is insignificantly different from zero after controlling for $r_{i,t-1}^{PS}$ (column 4). Overall, the results reported in panel B are like those in panel A and confirm our key findings of return predictability by text-based innovation similarity even after controlling for text-based industry momentum.

3.4 Return predictability over time

McLean and Pontiff (2016) show that return predictability of many discovered anomalies fades with the passage of time or postpublication of academic articles that document them. With their study as a backdrop, we investigate the robustness of the return predictability by technologically similar firms over time. Table 6 presents the results for four subsamples: 1977–1986

Table 6
Return predictability by technologically similar firms over time

A: Return predictability by text-based technologically similar peers

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$
$r_{i,t}^{TS-text}$	0.916***	1.163***	0.726***	0.939***	0.905***
	(8.162)	(5.044)	(4.053)	(5.341)	(3.729)
$r_{i,t}^{Ind}$	0.124**	0.122	0.171*	0.198*	−0.042
	(2.321)	(1.081)	(1.850)	(1.711)	(−0.619)
$r_{i,t}$	−0.172*	−0.139	−0.016	−0.315	−0.313
	(−1.760)	(−0.531)	(−0.170)	(−1.529)	(−1.485)
$r_{i,t-12,t-1}$	0.235**	0.663***	0.455***	−0.164	−0.211
	(2.117)	(3.212)	(3.529)	(−0.936)	(−0.720)
LogBM	0.091	0.150	0.330	−0.031	−0.253
	(0.737)	(0.646)	(1.277)	(−0.235)	(−1.204)
LogSize	−0.081*	−0.061	−0.033	−0.141	−0.116
	(−1.670)	(−0.680)	(−0.379)	(−1.295)	(−1.650)
Constant	1.754***	2.126**	1.282	1.569	2.326***
	(3.413)	(2.075)	(1.450)	(1.290)	(3.082)
Obs. (N)	290572	19944	84885	106389	79354
R^2	0.082	0.145	0.064	0.066	0.051
Sample	Full sample	1977–1986	1987–1999	2000–2008	2009–2016

B: Return predictability by class-based technologically similar peers

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$
$r_{i,t}^{TS-class}$	0.735***	1.146***	1.014***	0.376	0.163
	(4.106)	(3.697)	(3.050)	(1.406)	(0.524)
$r_{i,t}^{Ind}$	0.183***	0.201	0.183**	0.278**	0.046
	(3.417)	(1.647)	(2.132)	(2.390)	(0.567)
$r_{i,t}$	−0.172*	−0.132	−0.037	−0.303	−0.301
	(−1.790)	(−0.527)	(−0.391)	(−1.437)	(−1.446)
$r_{i,t-12,t-1}$	0.236**	0.665***	0.430***	−0.144	−0.191
	(2.140)	(3.102)	(3.622)	(−0.807)	(−0.648)
LogBM	0.110	0.140	0.404	−0.048	−0.250
	(0.899)	(0.611)	(1.623)	(−0.351)	(−1.230)
LogSize	−0.084*	−0.076	−0.027	−0.145	−0.121*
	(−1.737)	(−0.836)	(−0.318)	(−1.327)	(−1.671)
Constant	1.779***	2.428**	1.197	1.458	2.353***
	(3.441)	(2.310)	(1.316)	(1.272)	(2.805)
Obs. (N)	284351	19476	83648	103992	77235
R^2	0.084	0.149	0.067	0.068	0.052
Sample	Full sample	1977–1986	1987–1999	2000–2008	2009–2016

C: Return predictability by "pure" text-based technologically similar peers

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$
$r_{i,t}^{TS-text\ \ NOT\ \ TS-class}$	0.270***	−0.039	0.111	0.583***	0.561***
	(4.276)	(−0.409)	(1.540)	(7.160)	(4.655)
$r_{i,t}^{TS-text\ \ AND\ \ TS-class}$	0.680***	0.823***	0.665***	0.606***	0.615***
	(7.720)	(2.858)	(6.768)	(5.796)	(3.660)
$r_{i,t}^{TS-class\ \ NOT\ \ TS-text}$	0.173	0.399	0.519*	−0.112	−0.372
	(0.994)	(0.947)	(1.745)	(−0.462)	(−1.652)
$r_{i,t}^{Ind}$	0.108**	0.071	0.146	0.188*	−0.005
	(2.037)	(0.610)	(1.578)	(1.684)	(−0.070)
$r_{i,t}$	−0.151	−0.063	−0.004	−0.326	−0.309
	(−1.535)	(−0.232)	(−0.045)	(−1.601)	(−1.557)

(continued)

Table 6
Continued

C: Return predictability by "pure" text-based technologically similar peers

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$
$r_{i,t-12,t-1}$	0.247**	0.699***	0.436***	−0.137	−0.192
	(2.184)	(2.964)	(3.649)	(−0.765)	(−0.673)
LogBM	0.159	0.322	0.392	−0.015	−0.243
	(1.312)	(1.415)	(1.596)	(−0.109)	(−1.198)
LogSize	−0.077	−0.061	−0.017	−0.139	−0.126*
	(−1.572)	(−0.651)	(−0.195)	(−1.276)	(−1.834)
Constant	1.649***	2.100*	0.959	1.552	2.390***
	(3.190)	(1.941)	(1.060)	(1.380)	(2.988)
Obs. (N)	274487	17455	79217	101362	76453
R^2	0.096	0.179	0.074	0.072	0.057
Sample	Full sample	1977–1986	1987–1999	2000–2008	2009–2016

This table provides the results of return predictability Fama-MacBeth regression by technologically similar firms over time. The main predictive variables are the portfolio return of text-based technologically similar peers (in panel A), the portfolio return of class-based technologically similar peers (in panel B), and the portfolio return of "pure" text-based technologically similar peers (in panel C). Pure text-based technologically similar peers are those that do not overlap with class-based peers. All other independent variables are the same as in Table 5 and defined in the Appendix. All returns are in excess of the risk-free rate and are standardized to have a standard deviation of one. t-statistics are reported in parentheses and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. The symbols ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

(column 2), 1987–1999 (column 3), 2000–2008 (column 4), and 2009–2016 (column 5).

In Panel A, we report the performance of our novel text-based innovation similarity. We find that text-based return predictability is active in all subperiods as evidenced by positive and significant coefficient estimates on $r_{i,t}^{TS-text}$: 1.163, 0.726, 0.939, and 0.905 in columns 2, 3, 4, and 5, respectively.

Panel B present the results of predictive Fama and MacBeth regressions of class-based similar portfolios. In contrast to the text-based innovation similarity, we find that predictability by patent class-based technologically similar firms is not active after the year 2000 in our sample of firms with stock prices greater than \$5. Indeed, the regression coefficients on $r_{i,t}^{TS-class}$ during 2000–2008 and 2009–2016 are not statistically significant.[19]

In Panel C of Table 6, we assess the performance of the nonoverlapping ("pure") text-based portfolio $r_{i,t}^{TS-text\ NOT\ TS-class}$, which is the average monthly return of firms that are text-based technologically similar, but not class-based similar, to a focal firm. According to the regression estimates, before 2000, "pure" text-based portfolios do not predict stock returns. However, the portfolios exhibit strong stock return predictability during 2000–2008 and 2009–2016 as shown by the significant and positive estimates

[19] Using a sample of firms with stock prices above \$1, LSWZ (2019) find return predictability during their whole sample period with markedly smaller alphas during 2000–2012. We were able to replicate their results. Nevertheless, when we extend the sample to 2016, return predictability is no longer active in the 2009–2016 sample period (even for stocks above \$1).

on $r_{i,t}^{TS-text\ \ NOT\ \ TS-class}$ (0.583 and 0.561 in columns 4 and 5, respectively). These results underscore the economic importance of the new information contained in the "pure" text-based technological links that cannot be discovered through the traditional patent-class taxonomy. The trading strategy of buying (selling) stocks that earned high (low) returns during the previous month on their "pure" text-linked peers yields an average monthly return of 58.3 and 56.1 basis points during the 2000–2008 and 2009–2016 periods, respectively.

The predictability pattern of "pure" text-based firms highlights how investors' practice of incorporating patent information has evolved. Patents' text was not available online until 1999. Without digital access, direct patent text examination was most likely a "privilege" afforded to specialists doing research in the field.[20] In 1996, Congress amended the FOIA to require agencies to create electronic reading rooms for records that were more likely to be subject to FOIA requests. As a result, starting in 1999, the public was able to access images of all patents issued from 1976 to the present. We conjecture that, with the advent of online access, anyone could examine patent text and draw conclusions about firms' technological positions. In line with this conjecture, we find that "pure" text-based peers exhibit strong statistical and economical predictability from 2000 onwards.

We argue that during the subperiods before 1999, the unavailability of internet access to patent documents led to the absence of 1-month ahead return predictability of the pure text-linked firms (panel C of Table 6). However, costly information acquisition should not eliminate stock return predictability. Instead, it is possible that before 1999 higher cost of information acquisition related to technologically similar patents could delay the incorporation of this information into stock prices. To test this conjecture, we alter the specification in panel C of Table 6 by changing (a) the time period to 1977–1999 and (b) the dependent variable to be the 6-month cumulative excess return. The estimates of the resulting regression are as follows:

$$r_{i,t+1\rightarrow t+6} = \underset{(1.824)}{0.683}\ r_{i,t}^{TS-text\ \ NOT\ \ TS-class} + \underset{(3.450)}{1.842}\ r_{i,t}^{TS-text\ \ AND\ \ TS-class} + \underset{(0.793)}{0.818}\ r_{i,t}^{TS-class\ \ NOT\ \ TS-text} + C + e_{i,t}.$$

The positive and significant estimate on $r_{i,t}^{TS-text\ \ NOT\ \ TS-class}$ (t-statistics reported in parentheses) indicates that, prior to 1999, "pure" text-based portfolios predict stock returns over a longer horizon.

We are sensitive to the concern that the division into four different time periods in Table 6 is somewhat random, as the lengths of the subperiods—ranging from 8 to 13 years—are not uniform. To address this concern and

[20] Interested public still could read patents; however, extra steps were needed, such as accessing paper archives or microfilm storage.

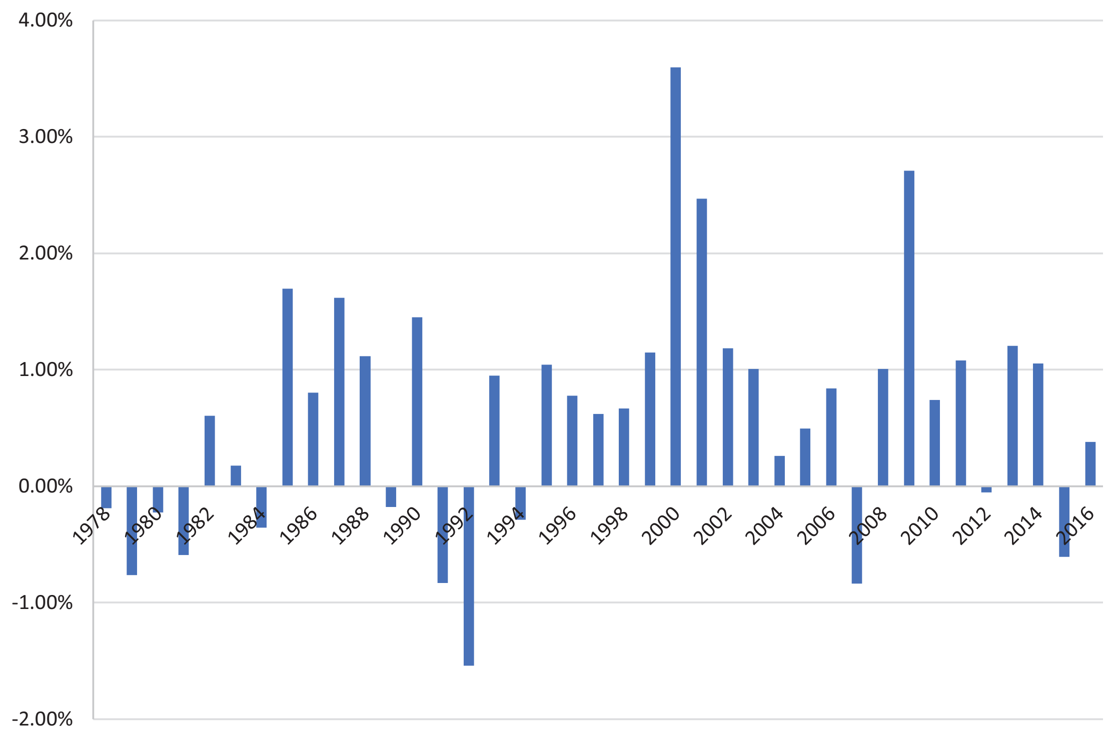

Figure 2
Average alpha of self-financing trading strategy by calendar year
This figure reports average monthly alphas by calendar year of the self-financing trading strategy consisting of buying high portfolios and shorting low equal-weighted portfolios. These portfolios are sorted into deciles at the beginning of each month based on return of "pure" text-based technologically similar firms (those firms that have technological connections identified via textual analysis and without patents in the same patent class) in the previous month. Alpha is the intercept in a regression of monthly excess return on factor returns from the Fama-French three-factor model, which includes MKT (the market factor), SMB (the size factor), and HML (the value factor).

show the stability of the cross-predictability we uncover, Figure 2 plots the average alpha from a self-financing long-short strategy on a year-by-year basis. The pattern illustrated in Figure 2 supports our inferences from the results in Table 6 that, after 2000, text-based peers exhibit strong statistically and economically meaningful predictability.

Overall, the findings reported in this section illuminate important differences between our work and LSWZ (2019). Future research could build on our work because, unlike LSWZ, the stock return predictability anomaly related to our text-based measure is both active and strong.

3.5 Underlying mechanisms of return predictability

Our results suggest that return predictability by technologically similar firms is more likely driven by slow information diffusion due to investors' limited processing capacity or inattention. The limited-information hypothesis predicts that more obscure information is more likely to be overlooked by investors and incorporated more slowly into stock prices. Thus, the magnitude of return cross-predictability should be higher when the economically important information thought to be overlooked is less obvious or more complex. In

Table 7
Effect of information environment on predictability

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$	(5) $r_{i,t+1}$
$r_{i,t}^{TS-text}$	0.916***	0.725***	0.701***	1.026***	1.029***
	(8.162)	(5.507)	(6.142)	(7.312)	(6.899)
$r_{i,t}^{TS-text} * High\ Complexity$		0.374*			
		(1.827)			
$r_{i,t}^{TS-text} * High\ Exploitative$			0.482**		
			(2.375)		
$r_{i,t}^{TS-text} * High\ Explorative$				−0.178	
				(−0.968)	
$r_{i,t}^{TS-text} * High\ Common\ Analyst$					−0.416***
					(−2.666)
$High\ Complexity$		−0.099			
		(−0.735)			
$High\ Exploitative$			0.026		
			(0.187)		
$High\ Explorative$				0.007	
				(0.048)	
$High\ Common\ Analyst$					0.004
					(0.044)
$r_{i,t}^{IND}$	0.124**	0.111**	0.126**	0.120**	0.097*
	(2.321)	(2.165)	(2.321)	(2.342)	(1.728)
$r_{i,t}$	−0.172*	−0.171*	−0.180*	−0.175*	−0.254**
	(−1.760)	(−1.740)	(−1.811)	(−1.790)	(−2.463)
$r_{i,t-12,t-1}$	0.235**	0.255**	0.236**	0.242**	0.213*
	(2.117)	(2.270)	(2.117)	(2.176)	(1.650)
BM	0.091	0.120	0.093	0.077	0.178
	(0.737)	(1.007)	(0.766)	(0.625)	(1.061)
$LogMV$	−0.081*	−0.077	−0.086*	−0.085*	−0.167***
	(−1.670)	(−1.628)	(−1.778)	(−1.761)	(−3.121)
Constant	1.754***	1.737***	1.754***	1.810***	2.476***
	(3.413)	(3.460)	(3.499)	(3.580)	(4.494)
Obs. (N)	290,572	290,572	290,572	290,572	240,145
R^2	0.082	0.091	0.089	0.089	0.094

This table reports results a series of cross-sectional analyses designed to evaluate the effect of firms' characteristics on return predictability by technologically similar peers. The sample period is 1977–2016. All tests consist of Fama-MacBeth predictive regressions where the dependent variable is stock returns of firm i at month $t+1$ ($r_{i,t+1}$). The explanatory variables are lagged portfolio returns of firm i's text-based technological peers ($r_{i,t}^{TS-text}$), industry portfolio returns ($r_{i,t}^{IND}$), firm i's stock returns in the previous month ($r_{i,t-1}$) and over the preceding 11 months ($r_{i,t-12,t-2}$), firm i's book-to-market ratio ($BM_{i,t}$) and a natural logarithm of market value ($LogMV_{i,t}$). In column 1, we present the baseline results from Table 5. In column 2, we interact $r_{i,t}^{TS-text}$ with the indicator variable *High_complexity*, which equals one if a firm's patent complexity is above year-median, and zero otherwise. In column 3, we interact $r_{i,t}^{TS-text}$ with the indicator variable *High Exploitative*, which equals one if a firm's ratio of technological exploitativeness is above year-median, and zero otherwise. In column 4, we interact $r_{i,t}^{TS-text}$ with the indicator variable *High Explorative*, which equals one if a firm's ratio of technological explorativeness is above year-median, and zero otherwise. In column 5, we interact $r_{i,t}^{TS-text}$ with the indicator variable *High Common Analysts*, which equals one if the percentage of common analysts relative to the total number of analysts covering a firm is above the year-median, and zero otherwise. A common analyst is an analyst who issues more than one earnings-per-share forecast per year for both firms in a technological pair. A firm's patent complexity, technological exploitativeness and explorativeness, together with other independent variables, are defined in the Appendix. All return variables are in excess of the risk-free rate and are standardized to have a standard deviation of one. t-statistics are reported in parentheses and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7, we investigate these issues with five different return predictability Fama and MacBeth regressions. For reference, column 1 provides the baseline result reported in column 1 of Table 5.

3.5.1 Complex innovation. Using our Big Data approach, we develop several constructs to describe a firm's innovation space and strategy. First, we consider a firm's innovation complexity, which is defined as the average number of keyphrases in a firm's patents. The intuition is that the more keyphrases a patent contains the more "complex" it is. A keyphrase is a technical term as defined by human experts. Keyphrases are taken from the titles or subtitles of Wikipedia entries. Note that the number of keyphrases is not a function of patent length (the number of words). Some patents can be shorter but filled with keyphrases that might get very technical, while other patents might be longer but consist of more-general (and potentially less-complex) text with fewer keyphrases.[21]

Column 2 of Table 7 reports the coefficient estimates of the baseline regression augmented by the interaction of $r_{i,t}^{TS-text}$ with *High Complexity*, which is a dummy variable that equals one if a firm's innovation complexity is above the year-median for that variable, and zero otherwise. The positive and statistically significant coefficient estimate on the interaction term (0.374) indicates that firms with high innovation complexity exhibit stronger return predictability. This result supports the limited-processing/inattention hypothesis.

3.5.2 Exploitative and explorative innovation. Another dimension that characterizes innovation strategy is the degree to which a firm's patents are exploitative or exploratory. Leveraging our textual analysis, we extend the exploitativeness and explorativeness measures used in other studies (see, for example, Brav et al. 2018). A patent is considered exploitative if at least 80% of its text-based technological connections are with the same firm's existing knowledge—that is, its own patents. In contrast, a patent is exploratory if most of the technological space is occupied by knowledge that is new to the firm. Thus, a patent is defined as explorative if at least 80% of its text-based technological links are not with its own patents. Aggregated at the firm-year level, the percentage of exploitative/exploratory patents indicates whether a firm's innovative strategy relies on existing knowledge (e.g., incremental advances relative to existing patents) or focuses on exploring new technologies.[22] The effects of exploitativeness and explorativeness on return predictability appears in columns 3 and 4 of Table 7, respectively. The estimate on the interaction term of $r_{i,t}^{TS-text}$ and *High Exploitative*—a dummy variable that equals one if the ratio of a firm's innovation exploitativeness is above the year-median, and zero otherwise—is positive and significant (0.482). This result indicates that investors have difficulty incorporating information about

[21] As reported in Table 1, panel A, a firm's technological complexity has a mean of 0.05 and a standard deviation of 0.05. These values indicate considerable cross-sectional variation in the firm's technological complexity.

[22] As reported in Table 1, panel A, the means of exploitative and explorative patents are 0.15 and 0.09, respectively. They also exhibit a standard deviation of 0.17 and 0.11, respectively.

innovation-related fundamental news of the focal firm's technological neighbors if most of its links are with its own technology. The intuition is that if most of a firm's patent connections are "internal," the information flow related to its innovation activity is limited. Therefore, the spillover effect is less obvious for market participants whenever a firm's innovations are built upon (exploit) its previous inventions.

Column 4 of Table 7 shows that a firm's explorative innovation strategy does not have an effect on the return predictability as evidenced by the insignificant coefficient on the interaction of $r_{i,t}^{TS-text}$ and *High Explorative*. Indeed, the effect of the explorativeness is less clear. On the one hand, if most of the technological connections of a firm are "external," the information flow is unrestricted. On the other hand, if a firm explores new frontiers and recombines knowledge from many different fields, it could be difficult for investors to understand is technological position. These two channels can counterbalance each other, leading to no effect on return predictability.

3.5.3 Analyst coverage. Limited information models also predict that the magnitude of return cross-predictability should be lower when more investors are informed about the potentially overlooked information. We use the number of analysts who cover both firms in a technological pair as a proxy for investors who are informed about technological links and pay attention to the information flow. Financial analysts are important information intermediaries who gather, analyze, and disseminate relevant information about firms and industries (Hong, Lim, and Stein 2000; Gleason and Lee 2003). Tan, Wang, and Yao (2017) show that analysts exploit information spillover between technologically similar firms to increase their forecast accuracy and lower the cost of information acquisition. Thus, analysts who cover both firms in a technologically linked pair are likely to be aware of their innovation similarity and to inform market participants that connected firms are affected by the same economic shocks. To evaluate this conjecture, we use analyst forecast data from the Institutional Brokers' Estimate System (I/B/E/S) detailed history file and consider forecasts of earnings per share, which are the most commonly available type of forecast. We define "common analyst" as any analyst who issues more than one earnings-per-share forecast per year for both firms in a pair.

Column 5 of Table 7 presents the results of the predictive regression augmented by the interaction term of $r_{i,t}^{TS-text}$ with *High Common Analyst*, which is a dummy variable that equals one if the number of common analysts scaled by the total number of analysts covering the focal firm is above the year-median for this variable, and zero otherwise. The estimate for the interaction term is negative and statistically significant (−0.416), suggesting that firms with more common analysts exhibit lower return predictability and providing support for the limited-processing/inattention hypothesis.

Table 8
Robustness tests

	(1) $r_{i,t+1}$	(2) $r_{i,t+1}$	(3) $r_{i,t+1}$	(4) $r_{i,t+1}$
$r_{i,t}^{TS-text}$	0.916***	0.749***	0.399***	0.899***
	(8.162)	(6.546)	(3.418)	(7.761)
$r_{i,t}^{CUST}$				0.532*
				(1.857)
$r_{i,t}^{SUP}$				−0.039
				(−0.101)
$r_{i,t}^{Ind}$	0.124**	0.148***	0.194***	0.059
	(2.321)	(2.787)	(2.727)	(1.098)
$r_{i,t}$	−0.172*	−0.162	−0.279**	−0.207**
	(−1.760)	(−1.546)	(−2.545)	(−2.040)
$r_{i,t-12,t-1}$	0.235**	0.272**	0.228	0.241**
	(2.117)	(2.138)	(1.560)	(2.041)
$LogBM$	0.091	0.137	0.094	0.172
	(0.737)	(0.965)	(0.686)	(1.397)
$LogMV$	−0.081*	−0.074*	−0.121**	−0.095*
	(−1.670)	(−1.752)	(−2.458)	(−1.905)
Intercept	1.754***	1.678***	2.125***	1.414**
	(3.413)	(3.656)	(4.080)	(2.109)
Obs. (N)	290572	215043	142949	246003
Adj. R^2	0.082	0.096	0.118	0.096
Sample restriction	Full sample	Exclude firms with market cap less than 20th pct. NYSE	Excludes firms with market value greater than the focus firm's	Full sample

This table presents robustness analyses of return predictability results. The sample period is 1977–2016. The tests consist of Fama-MacBeth predictive regressions where the dependent variable is the stock return of focal firm i in month $t+1$. Column 1 provides the baseline result reported in the Table 5 for the full sample. Column 2 shows the results for a sample of firms with a market value above the 20th percentile of NYSE market capitalization. In column 3, we drop all links when peer firms are larger than focus firm i. In column 4, we augment the model with two new predictive variables $r_{i,t}^{CUST}$ and $r_{i,t}^{SUP}$, which are portfolios of customer and supplier industry returns based on the Bureau of Economic Analysis's (BEA's) input–output tables (Menzly and Ozbas 2010), respectively. All other independent variables are the same as in Table 5 and defined in the Appendix. All return variables are in excess of the risk-free rate and are standardized to have a standard deviation of one. t-statistics are reported in parentheses and the standard errors are Newey-West adjusted (up to 12 lags) for heteroskedasticity and autocorrelation. The symbols ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

4. Robustness Tests

We perform three robustness tests and report the findings in Table 8. To facilitate comparisons, column 1 in Table 8 provides the baseline result from column 1 of Table 5.

To address the possibility that a thin market might be driving the cross-predictability results, column 2 excludes stocks with market capitalization below the 20th percentile of the NYSE. The estimate on $r_{i,t-1}^{TS-text}$ remains statistically significant (0.749). In column 3 we drop all links when a peer firm is larger than a focus firm i to ensure that our results are not driven by the lead–lag effect of big and small firms (Hou 2007). The estimate on $r_{i,t-1}^{TS-text}$ becomes smaller (0.399) but remains statistically significant. This decrease in cross-predictability is consistent with the information flow from large to small firms within an industry, as documented by Hou. Importantly, these findings

Table 9
Return predictability: the event study of patents issuance

	(1) Buy-and-hold abnormal returns during 3 days at patent grant date		(2) 1 month buy-and-hold returns that starts cumulating 2 days after patent grant date
Focal firms		**Focal firms**	
BH size and BM adj t(−1,+1)	5.93	*BH size and BM adj 1m*	0.05
	(243.24)		(0.74)
Peer firms: TS-text not industry		**Peer firms: TS-text not industry**	
BH size and BM adj t(−1, 1)	0.26	*BH size and BM adj 1 m*	0.23
	(45.95)		(14.22)

This table reports results of analyses of the market response of technologically connected peers to news about focal firms' patenting activity. In column 1, we show 3-day buy-and-hold size- and book-to-market-adjusted returns of a focal firm and its technological peers centered around a focal firm's patent issue date. Technological peers are determined via textual analysis of patents. We consider only technological peers that do not belong to the same industry as a focal firm. Column 2 displays subsequent monthly buy-and-hold returns that start cumulating 2 days after a patent grant date. We report *t*-statistics in parentheses.

indicate that information can flow from smaller to larger stocks when economic links are captured by technological connectedness.

Finally, to rule out the possibility that our results are driven by customer–supplier links (Menzly and Ozbas 2010; Cohen and Frazzini 2008), we include the portfolio return of customer and supplier industries, $r_{i,t-1}^{CUST}$ and $r_{i,t-1}^{SUP}$, respectively, in the predictive regression. The results, which appear in column 4, show that the estimate on $r_{i,t-1}^{TS-text}$ is not affected by the inclusion of these control variables.

To summarize, our results withstand a battery of robustness checks designed to ensure that the limited liquidity of small stocks, the predictability of small firms by large firms, and the effect of customer–supplier links do not drive our results. Overall, our findings suggest that limited processing/inattention drives the predictability results. We show that the return predictability of technologically connected firms is largely unexplained by well-known risk factors. The cross-sectional analysis results are consistent with the limited-processing/inattention hypothesis indicating that investors are slow in incorporating information about firms with more-complex innovations. Importantly, in subsection 2.3.3 we find real economic connections in terms of contemporaneous and predictive relations between the focal firm's ROA (R&D) and the ROA (R&D) of its technologically linked firms. These findings also cast doubt on the alternative that some unobservable risk might be driving our results.

5. Return Predictability from Patenting Activity of Focal Firms

The preceding tests indicate that the stock return of a focal firm predicts that of its text-based technologically linked peers. Yet, those tests do not unambiguously tell us what the information is that drives the initial stock return of the focal firm. We contend that the driver in question is the information about the focal firm's technologies common across firms. To shed light on this issue, we examine whether the stock market response to news about a focal firm's patent grant positively predicts subsequent stock returns of its peer firms that share similar technologies. To sharpen any inferences arising from this test, we isolate the effect of technological connectedness by only considering technological peers that do not belong to the focal firm's industry.

We match each patent in our sample to those in the Kogan et al. (2017) database to identify a patent's grant date and its assessed market value. Next, for every month, we rank firms into market value quintiles and focus on firms in the highest quintile under the rationale that the informational signal to the market is the strongest for such patent news. We report the initial and subsequent market reaction to a patent grant for focal firms and their peers in column 1 of Table 9. We observe a positive and statistically significant abnormal return for focal firms when a patent is granted. The 3-day size- and book-to-market-adjusted buy-and-hold $(BH\_size\_BM\_adj\_t(-1,+1))$ return centered at the issuance date equals 5.93%. Technological peers also earn significantly positive abnormal returns upon the focal firm's patent grant date. For these peers, the $(BH\_size\_BM\_adj\_t(-1,+1))$ return is 0.26%.

In column 2 of Table 9, we report the post-patent-grant-announcement market response measured by monthly buy-and-hold returns for focal firms and their technological peers. We start cumulating monthly buy-and-hold returns from the 2nd day after a patent issuance date. The size- and book-to-market-adjusted buy-and-hold return for a focal firm is not statistically significant. This result suggests that positive news about a patent grant is immediately priced for focal firms. By contrast, technological peers exhibit positive and statistically significant subsequent monthly returns. Indeed, for peer firms, we estimate a statistically significant $(BH\_size\_BM\_adj\_t(-1,+1))$ return equal to 0.23%. Together, the tests in Table 9 provide evidence suggesting that news about a focal firm's patents predicts subsequent returns for peer firms connected to the focal firm through shared technologies but not necessarily through shared industry membership.

6. Conclusions

Using the publicly available, voluminous, and extremely rich data in patent texts, we construct a novel measure of technological similarity between firms. In contrast to the traditional innovation-distance proxies based on a patent's technological class as assigned by USPTO or on common citations, our measure avoids biases induced by the discretion of patent examiners and patent attorneys and by firms' citing strategies and practices. Importantly, our text-based measure of innovation similarity captures important new technological links that cannot be discovered using traditional measures.

We use our text-based measure to study how information about technological proximity is incorporated into stock prices. Our analyses generate robust empirical evidence showing that information about innovation links is impounded into stock prices as evidenced by contemporaneous comovement of technologically similar stocks. Yet, as suggested by the return predictability for technologically similar firms, information about innovation links is not quickly or fully impounded into stock prices. The estimated effects are of first-order economic importance: a self-financing trading strategy of buying technologically linked firms whose peers earned high returns in the previous month and selling firms whose peers earned low returns in the previous month yields a 17% annual premium.

The analyses show stronger predictability for firms with higher fractions of either exploitative or complex patents. At the same time, we also find weaker predictability for firms with a larger number of analysts who also provide coverage for their technological neighbors. These findings, which illuminate potential channels underlying our baseline results, support the limited-information-processing capacity hypothesis to explain the technology momentum.

Overall, our findings indicate that (1) intangible information about innovation links is not efficiently impounded into stock prices and (2) unlike the evidence from work that uses USPTO patent classes to establish technological similarity, the Big Data text-based return predictability we uncover is still active. These results underscore the importance of processing massive and complex data related to innovation activity in the new knowledge-based economy.

Appendix. Variable Definition

Technological and product similarity	
$TS\text{-}text_{ij}$	The text-based technological similarity between firms i and j assessed through textual analysis of firms' patents. $TS\text{-}text_{ij}$ is computed as a natural log of the sum of the text-based technological similarity between patents of firms i and j discounted by the patent age and normalized by the log of the product of the total number of patents for each firm in a pair. We discount each pair by the discounting factor $1 - \tau/T$, where τ is the age of the newer patent in the pair in a given year and T is the depreciation period, which equals 20 years (the legal life of a patent). The similarity between a pair of patents is assessed using two MapReduce tasks. The first MapReduced phase looks up candidate professional terms in patents (the Map phase). The second phase (the Reduce phase) produces the list of patents that contains each detected term filtering out lists that are too long and correspond to many common terms. The second MapReduced task computes the distance between the two patents, represented as a vector of their common terms weighted by a TF-IDF (term frequency-inverse document frequency) variant (Manning and Schütze 1999).
Indicator $TS\text{-}text_{ij}$	Value equals one if firms i and j are text-based technologically similar, and zero otherwise.
$TS\text{-}class_{ij}$	The class-based technological similarity between firms i and j computed by the Jaffe method (Jaffe 1986) as the cosine similarity between the two firms' innovation profile vectors $\frac{\sum_k p_{1k}p_{2k}}{\sqrt{\sum_k p_{1k}^2 \sum_k p_{2k}^2}}$, where $p1$ and $p2$ are innovation profiles of firms i and j, respectively. Innovation profile is a vector, $p = (p_1, p_2, \ldots, p_K)$, across K dimensions, where p_k equals the fraction of patents in the k^{th} patent class held by the firm (Ahuja 2000; Song, Almeida, and Wu 2003).
Indicator $TS\text{-}class_{ij}$	Value equals one if firms i and j are class-based technologically similar, and zero otherwise.
Indicator $TS\text{-}text\text{-}NOT\text{-}class_{ij}$	Value equals one if firms i and j are text-based technologically similar but are not class-based technologically similar, and zero otherwise.
$TS\text{-}cite_{ij}$	The cite-based technological similarity between firms i and j computed as the sum of number of citations between firms i and j discounted by the patent age and normalized by the log of the product of the total number of patents of each firm in a pair. We discount each pair by the discounting factor $1 - \tau/T$, where τ is the age of the newer patent in the pair in a given year and T is the depreciation period, which equals 20 years (the legal life of a patent).
Indicator $TS\text{-}cite_{ij}$	Value equals one if firms i and j cite each other's patents, and zero otherwise.
PS_{ij}	The product similarity based on the dynamic TNIC classification constructed using text-based analysis of the firms' product description in the 10-K filing. TNIC classification is developed in Hoberg and Phillips (2010, 2016) and available at https://hoberg-phillips.tuck.dartmouth.edu/industryclass.htm.
Indicator PS_{ij}	Equals one if firms I and j are in the same dynamic TNIC industry and zero otherwise.
Portfolio returns of technologically and product similar firms	
$r_{i,t}^{TS-text}$	The monthly portfolio returns of the text-based technologically similar firms computed as the average monthly returns of the firms that belong to firm i's text-based technological space, weighted by the strength of their text-based similarity to firm i.
$r_{i,t}^{TS-class}$	The monthly portfolio returns of the class-based technologically similar firms computed as the average monthly returns that belong to firm i's class-based technological space, weighted by the strength of their class-based similarity to firm i.

(continued)

Continued

Technological and product similarity	
$r_{i,t}^{TS-text\ NOT\ TS-class}$	The monthly portfolio returns of firms that are text-based technologically similar but not class- similar to firm *i*, weighted by the strength of text-based technological similarity.
$r_{i,t}^{TS-cite}$	The monthly portfolio returns of citation-based technologically similar firms computed as the average monthly returns that belong to firm *i*'s citation-based technological space, weighted by the strength of their citation-based similarity to firm *i*.
$r_{i,t}^{Ind}$	The monthly value-weighted average portfolio returns of firms that belong to the same Fama-French 49 industry classification as firm *i*.
$r_{i,t}^{PS}$	The monthly portfolio returns of product similar firms computed as the average monthly returns that belong to firm *i*'s dynamic TNIC industry (Hoberg and Phillips 2010, 2016), weighted by the strength of the product similarity.

Stock returns and firms' characteristics	
$r_{i,t}$	The stock return of firm *i* in month *t* adjusted for the risk-free return.
$r_{i,t-12,t-2}$	The average return of firm *i* over the preceding 11 months starting with month *t*–12 (all returns are monthly to maintain comparability across the stock return variables).
$LogMV_{i,t}$	The natural logarithm of firm *i*'s market value computed as the product of shares outstanding and stock price at the end of fiscal year *t*.
$BM_{i,t}$	The book-to-market ratio calculated as the ratio of the book value of firm *i's* equity and its market value at the end of fiscal year *t*.
$LogAT_{i,t}$	The natural logarithm of total assets.
$ROA_{i,t}$	Return on assets of firm *i* at year *t*, computed as the earnings before interest, taxes, depreciation, and amortization scaled by total assets.
$Leverage_{i,t}$	Total debt of firm *i* in year *t* scaled by total assets.
$Cash_{i,t}$	Cash and short-term investments of firm *i* in year *t* scaled by total assets.
$RD_{i,t}$	Research and development expenses scaled by total assets.
$Number\ of\ patents_{i,t}$	The number of firm *i*'s patents filed that were approved by the USPTO in the past 20 years.
$Complexity_{i,t}$	The average number of keyphrases in firm *i*'s patents in year *t* normalized by the maximum complexity level.
$Exploitative_{i,t}$	The ratio of firm *i*'s exploitative patents to the total number of patents in year *t*. A patent is exploitative if at least 80% of text-based technologically similar patents belong to its firm.
$Explorative_{i,t}$	The ratio of a firm *i*'s explorative patents to the total number of patents in year *t*. A patent is explorative if at least 80% of text-based technologically similar patents do not belong to its firm.
$Diversity_{i,t}$	The average size of patent clusters in firm *i*'s patent portfolio for year *t*, computed as the total number of patents that a firm publishes before a given year, divided by the number of connected components in the graph of patents. The graph of patents is constructed for each firm and each year, where the nodes are patents published prior to or in that year and edges capture the semantic closeness between the patents: two patents are connected with an edge if they are semantically similar, and not connected otherwise.
$High\ Common_{i,t}\ Analyst_{i,t}$	Value equals one if for firm *i* and its technologically similar peers the number of common analysts is above the median in year *t*, and zero otherwise. A common analyst is an analyst who issues more than one earnings-per-share forecast per year for both firms in a technological pair.
$NumAnalysts_{i,t}$	The number of analysts who provide coverage for firm *i* in year *t*.

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